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Enhancing market return forecasts with an incident-based ESG indicator☆

Zhiyong Li [a], Weiping Qin [b,c,*]

[a] Albert Gubay Business School, Hen Goleg, Bangor University, Bangor LL57 2DG, United Kingdom
[b] Global Business School for Health, University College London, 7 Sidings Street, London, E20 2AE, United Kingdom
[c] Alliance Manchester Business School, University of Manchester, United Kingdom

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ABSTRACT

We introduce a novel incident-based ESG indicator that exhibits significant and robust time-series predictive power for aggregate equity market returns through in-sample and out-of-sample analyses across various contexts. This finding contrasts with most prior research, which primarily examines the cross-sectional relationship between ESG ratings and firm-level stock returns. We also show that a risk-averse investor can achieve substantially higher returns than the benchmark by applying our incident-based ESG market indicator. Additionally, we show that ESG incidents predict stock returns through both cash flow and discount rate news channels. We show that the predictive strength of our ESG market indicator likely arises from its ability to anticipate future market volatility and underlying firm fundamentals.

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1. Introduction

Environmental, social, and governance (ESG) investing is pivotal in today's investment landscape. Investors recognize that, especially in the context of stricter governmental ESG policies, examining firms' sustainable operations can safeguard their long-term interests. Furthermore, ESG investing contributes to a healthier environment and a stable economic system, benefiting everyone in the broader community. While the influence of ESG on firm performance and stock returns has been widely debated,[1] its predictive power on aggregate equity markets, a key asset pricing topic, has received less attention. Analyzing ESG issues at the market level offers valuable insights into their aggregate impact on both the overall market and broader economic performance and sustainability (Hansen, 2022). Aggregate ESG issues reflect common risks that affect the entire market. Pástor et al. (2021) argue that market-level greenness and investors' preferences for green assets affect the market equity premium. Thus, research on the market predictability of the ESG factor offers guidance for sustainable investment at the market level. Understanding how ESG factors influence the market is crucial for developing policies and regulations, especially when policymakers worldwide face significant monetary and fiscal challenges related to climate change (Faria et al., 2022).

This paper is among the first to study whether and how incident-based ESG information predicts future excess market returns in the U.S. stock market. We choose to focus our primary analysis on the U.S. stock market, following standard practice in foundational asset pricing research. As the world's largest and most liquid equity market, the S&P 500 serves as a crucial global benchmark, and establishing a new predictive relationship in this market is an important first step. Findings from the U.S. market can serve as a baseline for future international

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* Corresponding author at: Global Business School for Health, University College London, 7 Sidings Street, London, E20 2AE, United Kingdom.

E-mail addresses: z.li@bangor.ac.uk (Z. Li), weiping.qin@ucl.ac.uk (W. Qin).

[1] See the survey of Gillan et al. (2021), Khan (2022) and Atz et al. (2023).

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studies. Our study uses firm-level ESG incident data rather than ESG ratings to avoid the inconsistencies and issues associated with ratings.

First, we employ 28 ESG issues categorized by RepRisk to construct an ESG market indicator using the partial least squares (PLS) method, as detailed by Kelly and Pruitt (2015). The ESG market indicator demonstrates reliable forecasting power for the future excess returns of the S&P 500 index in the full sample analysis. This remains true even after controlling for various widely-used macroeconomic variables, the consumption cycle variable, and a series of uncertainty variables. In a single-predictor regression analysis, the ESG market indicator achieves a monthly R^2 of 7.902% and a coefficient of 1.327, which is significant at the 1% level. In two-predictor regression models, the coefficients of the ESG indicator under all settings are significant, ranging from 1.232 to 1.388. This suggests that other widely used predictors do not explain its robust predictive power for excess market returns.

To address concerns about overfitting in in-sample analysis, we extend our investigation to examine the out-of-sample predictive performance of incident-based ESG variables, thereby verifying our in-sample findings. We find that the aggregate ESG factor significantly improves the mean squared forecasting error (MSFE) by 14.420% relative to the historical average benchmark. The two-predictor performance with variables used in the in-sample analysis also consistently beats the benchmark in most cases. Therefore, the out-of-sample analysis further confirms the strong predictive power of our ESG market indicator in both single- and two-predictor settings.

To ascertain whether investors can benefit from predicting future excess market returns using incident-based ESG information, we also examine investors' asset allocation and evaluate the economic performance of an ESG-based investment strategy. Two measurements are employed: the certainty equivalent return (CER) gain and the Sharpe ratio. The results reveal that a risk-averse investor with a mean–variance utility function can achieve a CER 13.870% above the benchmark prediction and a Sharpe ratio of 1.306 by following forecasts of ESG incidents generated with the PLS method. The benefits can be even greater in two-predictor cases, where additional macroeconomic variables are incorporated alongside the ESG market indicator.

In addition, we explore the return predictability of four distinct ESG components: Environmental, Social, Governance, and Cross-cutting. We construct integrated indicators for each ESG component based on relevant individual issues. Our findings reveal statistically and economically significant return predictability for three of the components: Environmental, Social, and Cross-cutting. Notably, the social component has the most substantial influence on return forecasts. The model incorporating social information yields an out-of-sample R^2_{OS} of 9.607% at the 5% significance level, along with a CER gain of 13.288% and a Sharpe ratio of 1.276. However, it is important to note that these values are lower than those from the ESG market indicator. This suggests that the ESG market indicator has more robust predictive power than any individual integrated ESG component, which captures only a subset of the full ESG information.

Furthermore, we examine the forecasting performance of the ESG market indicator on different industry and factor portfolios. We collect 49 industry portfolio returns sourced from the Ken French data library and 9 factor portfolio returns as outlined by Carhart (1997), Fama and French (2015) and Hou et al. (2015). The results indicate that the ESG market indicator has significant predictive power for 47 of the 49 industry portfolios, except for the Precious Metals and Utilities industries. Moreover, the incident-based ESG variable can predict the investment factor portfolios separately described by Fama and French (2015) and Hou et al. (2015).

To test the potential impact of counter-cyclicality on the predictive power, we consider five measures of business cycles in regressions. The measures include recession dummies from the National Bureau of Economic Research (NBER) and the Organisation for Economic Co-operation and Development (OECD) (Atanasov et al., 2020) on consumption cycles, the Chicago Fed National Activity Index, and U.S. recession probabilities. Our analyses confirm that the ESG market indicator continues to demonstrate strong return predictability, significant at the 5% level. The impacts of business cycle variables on predictability vary based on the specific definition used. Notably, the NBER recession dummy and U.S. recession probability variables exhibit a counter-cyclical pattern, suggesting that the predictive power of the ESG market indicator increases during recession periods. The remaining three variables, however, do not show this pattern.

Based on valuation model theories, we study whether the predictive power stems from the cash flow channel, the discount rate channel, or a combination of both. Using the VAR framework outlined by Campbell (1991) and Campbell and Vuolteenaho (2004), we break down market returns into three components: expected returns, cash flow news, and discount rate news. Our findings indicate that the ESG market indicator predicts both cash flow and discount rate news, with the cash flow channel exhibiting significantly greater predictive strength. We further investigate whether the predictability arises from forecasting market volatility, firm fundamentals, and economic activity. The results suggest that the ESG market indicator does not forecast future economic activity indicators. On the other hand, the predictor can forecast future market volatility, firms' predictability, and valuation. Therefore, we can conclude that the ESG market indicator's predictability is partially attributable to its ability to forecast market volatility and firm fundamentals.

In additional robustness checks, we adjust various settings, and the results demonstrate that ESG-based investment strategies consistently yield significant economic benefits, even when accounting for higher transaction costs, higher risk aversion coefficients, and stringent portfolio weight constraints.

This paper contributes to the literature in several ways. First, while numerous studies have examined the contemporary relationship between ESG information and stock returns (Hsu et al. 2023, Li et al. 2023, Pástor et al. 2022, among others), and some have explored the predictive relationship, these studies remain at the firm level. The influence of ESG variables on future aggregate market

portfolios remains relatively unexplored. A few papers have theoretically highlighted the significant role of ESG factors in the aggregate market risk premium (Avramov et al., 2022; Azarmsa & Shapiro, 2023; Goldstein et al., 2022; Pedersen et al., 2021). The seminal work of Chu et al. (2022) empirically examines the predictive ability of ESG ratings for market returns over a short period from 2009 to 2019. However, ESG ratings are controversial due to inconsistencies among rating agencies and potential issues like 'greenwashing' and 'social washing'. We extend this line of research by examining the predictability of ESG information over the period from 2007 to 2021, including the COVID-19 period, thereby providing more comprehensive insights into the time-series predictive role of ESG information in asset pricing at the market level.

Second, to the best of our knowledge, we are the first to analyze the predictability of ESG incidents, rather than ratings, at the market level. To address concerns in the literature about inconsistencies among ESG rating vendors and the potential for 'greenwashing' and 'social washing', we advocate using incident-based ESG news as a solution in predictive analysis. Although a considerable number of papers have investigated the relationship between ESG variables and stock returns (Goldstein et al., 2022; Hong & Kacperczyk, 2009; Karolyi et al., 2023; Pástor et al., 2021, 2022), the evidence remains inconclusive due to divergent ESG scores across agencies that use different information sets and algorithms. While Glossner (2021) uses more objective incident-based ESG ratings, the algorithms behind these ratings remain a black box and may lag in reflecting the latest information. In contrast, using ESG-related events directly is more objective and timely than ratings, potentially overcoming these inconsistency issues.

Lastly, this paper also contributes to the sustainable investment literature and practices. Many papers estimate the performance of score-based ESG portfolio investment, such as Giese et al. (2019). Our study suggests that selecting individual stocks to form an ESG portfolio is unnecessary. Instead, a risk-averse investor can achieve benefits significantly higher than the benchmark by allocating assets between a risk-free asset and the market portfolio based on the forecasts from our incident-based ESG market indicator. Moreover, this investment approach is more cost-effective. It can be easily constructed using news feeds, thereby eliminating the need for additional subscriptions to various ESG score datasets. Allocating funds between the risk-free asset and the market portfolio is also more economical than maintaining a portfolio of individual stocks.

The remainder of the paper is structured as follows: Section 2 discusses the data and outlines the methodology for constructing the ESG market indicator. Section 3 presents the in-sample predictive results, while Section 4 evaluates the out-of-sample forecasting performance, considering both statistical and economic dimensions. Section 5 explores the potential explanations for our predictive findings. Section 6 details robustness tests, and Section 7 concludes the paper.

2. Data and ESG indicator construction

2.1. RepRisk incident data

RepRisk[2] screens daily for adverse ESG incidents and criticisms involving more than 300,000 public and private companies globally. This screening process leverages a combination of artificial intelligence and human intelligence to analyze over 150,000 public sources and stakeholders in 23 languages. These sources range from international, national, and local print and online media to social media, blogs, government bodies, regulators, NGOs, and think tanks. This broad scope ensures comprehensive coverage of publicly available information regarding ESG risks. This process ensures that our measure is less prone to managerial manipulation and reflects real-time, market-relevant information.

The RepRisk research process is driven by 28 distinct ESG issues (see Appendix Table A.1 for the full list), defined in accordance with key international standards such as the UN Global Compact and OECD Guidelines. Every identified adverse incident is linked to at least one of these issues, which fall into four primary categories: environmental, social, governance, and cross-cutting. This "issues- and event-driven" methodology allows RepRisk to capture any company exposed to ESG risks, rather than relying on a predefined company list.

For our study, we focus on incidents related to companies within the S&P 500 index. The raw data from RepRisk provide event-level records, and our process is designed to quantify the flow of this public information systematically. To illustrate, consider a news report detailing animal cruelty at a specific firm. An investor reading this article gains qualitative information about a potential risk. RepRisk's system simultaneously detects the article, classifies it under the 'Animal Mistreatment' issue, and records it as one incident. Thus, our data quantifies the same public event visible to market participants, enabling an objective, investor-relevant measure of ESG risk. Our methodology begins by counting these discrete, daily incident records for each of the 28 ESG issues across all S&P 500 firms. We then aggregate these daily counts to a monthly frequency to construct our final time-series variables, which represent the total volume of reported incidents for an issue in a given month.

Crucially, the RepRisk database is updated daily, capturing incident news as it becomes publicly available. Our methodology aggregates this daily data to form a monthly indicator at the end of month t, which is then used to forecast returns in month $t+1$. This construction ensures a clean out-of-sample forecasting exercise, as it relies solely on information available to an investor at the time the forecast is made and is consistent with studies using news-based indicators (Baker et al., 2016).

Our analysis uses the raw incident data provided by RepRisk, rather than derived scores such as the RepRisk Index (RRI) or other vendor ratings, ensuring that our

[2] For further details, please refer to www.reprisk.com and RepRisk's official methodology documentation https://www.reprisk.com/insights/resources/methodology#ii-research-approach-and-scope.

indicators are not affected by subsequent revisions to proprietary scoring algorithms. In constructing our predictor, we rely on the 28 issue categories that constitute the stable core of the RepRisk taxonomy, which have remained unchanged since 2007. These issue categories differ from the finer "topic tags" that RepRisk periodically updates to capture emerging themes, such as the "pandemics/epidemics" tag introduced in 2021. Because our variables are based on raw incidents aggregated at the issue level, later methodological or tagging adjustments do not influence our analysis.

2.2. Why do we use ESG incidents?

Instead of relying on ESG ratings, we recommend using ESG incidents directly to forecast stock returns, as an increasing number of ESG studies use ESG incidents rather than ratings as proxies for ESG information (e.g., Derrien et al. 2025, Glossner 2021). Popular ESG rating agencies such as Morningstar Sustainalytics, MSCI KLD, and Bloomberg ESG Rating employ their own distinct methodologies. Those are typically grounded in subjective analyst ratings, self-reported information, and firm commitments rather than actual outcomes based on incident occurrence (Berg et al. 2023, Li and Wu 2020). In recent years, it has become widely acknowledged that self-reported information is not a reliable data source, especially in the context of risk. Furthermore, the low correlation amongst ESG rating agencies casts doubt on the actual impact of ESG information provided by any single agency. The inconsistent ESG ratings may largely stem from the varied information sets and complex algorithms used by data vendors. The inconsistency leads to mixed results across various research topics. This issue can lead to a range of consequences, including higher costs of external financing, diminished predictive power for returns, reduced corporate motivation for ESG improvements, investor misunderstanding of ESG ratings, and the attenuation of bias in estimates (see, e.g., Avramov et al. 2022, Berg et al. 2023, 2022, Kim and Yoon 2023). Furthermore, ESG ratings, updated infrequently by data vendors, often lag in reflecting real-time market conditions, thus affecting investors' abilities to respond swiftly to market shocks and rapidly changing conditions (Li & Wu, 2020).

Using ESG incidents rather than ESG ratings stems from a basic tenet in finance: prices react to new information, and incidents are among the most direct and objective sources of information. RepRisk's methodology, which focuses on an "outside-in" approach by analyzing information from public sources and stakeholders while intentionally excluding company self-disclosures, aligns well with this tenet. As RepRisk states, "self-reported information is not reliable data – especially when it comes to risks".[3] This approach aims to provide a "reality check" on how companies conduct their business globally. Regardless of a firm's efforts (whether genuine or professed) to enhance its environmental responsibility, the outbreak of an incident suggesting that the firm is a major contributor to pollution can significantly impact its share price. This is due to potential doubts about the company's ability to efficiently uphold its environmental responsibility and the sincerity of its commitment. Such doubts can harm its reputation in the field. Therefore, we believe that outcome-based ESG performance proxies (incident-based in this context) are more straightforward and efficient predictors of stock returns. We primarily focus on actual negative ESG incidents, as they systematically reflect ESG risks and are typically sourced from public media and stakeholders rather than from direct company disclosures. Given that even positive ESG events can sometimes be influenced by company self-disclosures, we purposefully exclude them from our primary analysis. This approach helps mitigate inconsistencies in ESG ratings and the related issue of selective reporting.

By focusing on adverse events reported by third parties, our approach inherently mitigates concerns that firms can strategically time positive news or control the narrative. The timing of our predictor is driven by the public news cycle, not by corporate investor relations schedules. Therefore, while we acknowledge the general importance of considering strategic disclosure in ESG research, the risk of it overstating our specific predictive relationship is significantly reduced by the nature of our incident-based data. This concern is further minimized by recent findings showing that mandatory or third-party standards, rather than voluntary or firm-controlled disclosures, lead to more timely and consistent ESG information (Bochkay et al., 2025).

We collect the ESG data for individual companies listed on the S&P 500 from RepRisk via the Wharton Research Data Services (WRDS), covering the period from January 2007 to December 2021. This sample period encompasses several distinct economic regimes, including the Global Financial Crisis of 2008 and the unique market environment of the COVID-19 pandemic, which began in 2020. This rich historical variation allows for a robust evaluation of our predictive indicator's performance across different market conditions. Overall, our use of incident-based ESG data provides a timely, objective, and externally verified measure of ESG risks, well-suited for return forecasting.

2.3. Partial least square (PLS) method and ESG market indicator construction

Rapach et al. (2010) and Rapach and Zhou (2022) suggest that when forecasting with large information sets and noisy data, simply adding all the predictors into a single OLS regression is inappropriate. This is because such a regression becomes high-dimensional, leading to a substantial increase in the variance of the coefficient estimates. Furthermore, stock returns inherently contain a significant unpredictable component, which contributes to data noise. These challenges make OLS forecasting susceptible to in-sample overfitting, potentially leading to inadequate out-of-sample performance.

In this paper, we employ the PLS method, an effective approach that reduces the rank of the variables and extracts only one or a few factors from a large set of

[3] https://www.reprisk.com/insights/resources/methodology#ii-research-approach-and-scope

variables. This method was introduced to finance by Kelly and Pruitt (2015).[4]

First, we construct ESG issue-level indicators from monthly incident data. The primary input is the monthly aggregated number of adverse ESG incidents, denoted as $AC_{i,t}$. As explained in Section 2.1, this variable reflects the total number of reported adverse events for a given issue and month. We define each indicator, $esg_{i,t}$, as the difference between the current incident count and its six-month moving average. This approach is intended to capture unexpected increases or "shocks" in adverse activity. The underlying economic rationale is that deviations from recent history are more likely to convey new, market-relevant information than raw levels, which may reflect persistent trends or noise. By differencing against a moving average, our construction isolates the 'news' component of ESG incidents. Our findings are robust to this methodological choice: alternative constructions using a 12-month moving average or the raw $AC_{i,t}$ directly also produce statistically significant predictive power. Full results are available from the authors upon request.

The indicator is formally defined as follows:

$$esg_{i,t} = AC_{i,t} - \frac{1}{6}\sum_{j=t-6}^{t-1} esg_{i,j} \tag{1}$$

where, $AC_{i,t}$ is the aggregated counts of ESG incidents for issue i at month t, and $esg_{i,t}$ is the ith individual indicator at month t. Our raw monthly data on $AC_{i,t}$ for S&P 500 constituents spans from January 2007 to December 2021 (180 months). Due to the six-month lookback period required for calculating the moving average in Eq. (1), our resulting series for $esg_{i,t}$ covers the period from July 2007 to December 2021, yielding 174 monthly observations for each of the 28 individual indicators.

Summary statistics for these 28 individual indicators are presented in Table 1. The 'Obs' column in the table indicates these 174 monthly time-series observations. The volatility associated with individual indicators varies significantly. For instance, the "Animal mistreatment" indicator (IE1) exhibits the least variability, with a standard deviation of 4.976 and a range of −10.504 to 20.382. Conversely, the "Tax optimization" indicator (IG23) exhibits the highest volatility, with a standard deviation of 45.086. The maximum value for this issue (336.364) indicates that the incident count in a single month surpasses the average count over the preceding six months by 330.5.

Secondly, we construct the ESG market indicator using the PLS method. The details are described in Appendix C.

A crucial question is how our PLS-based indicator relates to a simpler, naive measure of aggregate incident activity. To visually examine the properties of our indicator and its relationship with raw ESG incident data, we present a detailed analysis in Fig. 1. The figure has been annotated with key historical events to provide relevant economic and political context.

In Panel A, we plot our PLS indicator alongside the simple sum of incidents, both in raw (non-detrended) form. The two series exhibit similar low-frequency trends, with a correlation of 0.535. This shared trend likely reflects the increasing prominence of ESG considerations during our sample period. The upward trajectory becomes especially noticeable after the Global Financial Crisis and intensifies around major policy milestones, such as the lead-up to the 2015 Paris Agreement. We interpret this trend as reflecting a real-world shift: a substantial increase in media and stakeholder reporting on ESG issues over time. As ESG gained prominence in public discourse, particularly following the Global Financial Crisis and during the build-up to the Paris Agreement, public scrutiny and coverage of corporate incidents appear to have intensified.

However, in time series forecasting, it is well known that shared trends between predictors and the dependent variable may lead to spurious regressions. To ensure that our in-sample results reflect a genuine month-to-month predictive relationship, we employ a detrended version of the PLS indicator for all in-sample analyses.

Panel B displays the two series after detrending. The visual co-movement largely disappears, and the correlation declines to 0.272. This result highlights the added value of the PLS methodology. It indicates that the predictive power of our indicator does not arise from trend co-movement. Instead, it reflects the PLS method's ability to extract meaningful information, acting as a "smart filter" for ESG signals, which is not captured by simple aggregation. Notably, the sharp spikes in the detrended PLS indicator often correspond to major global events. For instance, prominent spikes are observed around the 2017 United States presidential transition, a period of substantial policy uncertainty, and the onset of the COVID-19 pandemic in early 2020. These examples illustrate the indicator's responsiveness to systemic shocks. Understanding the economic nature of this non-obvious signal is the primary focus of our analysis in Section 5.

2.4. Stock market returns and other variables

This paper focuses on the U.S. stock market and calculates the monthly excess returns of the S&P 500 index by subtracting the U.S. Treasury bill rates from the log returns of the index.[5] The sample period spans from July 2007 to December 2021.

To assess the predictive power of our ESG index relative to other well-recognized variables, we use a set of previously explored predictors as our control variables. Initially, we incorporate the 14 macroeconomic variables established by Welch and Goyal (2008).[6] We also consider the consumption cycle variable, as described in Atanasov

[4] Kelly and Pruitt (2015) introduce the three-pass regression filter (3PRF) method and demonstrate that the PLS method is a special case of 3PRF. In this study, we adopt the PLS method using the 3PRF framework because the 3PRF approach to indicator construction is more economically intuitive. Specifically, our construction satisfies the following three conditions: "(i) the predictors are demeaned and variance-standardized in a preliminary step, (ii) the first two regression passes are run without constant terms, and (iii) proxies are automatically selected". Consequently, our method is effectively the PLS method.

[5] The data are available from https://sites.google.com/view/agoyal145.

[6] The data are available at https://sites.google.com/view/agoyal145.

Table 1
Summary statistics of issue-based ESG individual indicators.

Individual Indicators	Obs	Mean	Std.	Min	Max
Animal mistreatment (IE1)	174	2.499	4.976	−10.504	20.382
Climate change, GHG emission, and global pollution (IE2)	174	19.362	24.116	−33.550	127.008
Impacts on landscapes, ecosystems, and biodiversity (IE3)	174	34.465	26.733	−15.414	130.029
Local pollution (IE4)	174	26.611	21.908	−9.424	97.014
Overuse and wasting of resources (IE5)	174	2.990	5.988	−7.783	40.262
Waste issues (IE6)	174	9.230	11.079	−8.243	68.729
Child labor (IS7)	174	4.924	8.479	−7.141	45.651
Discrimination in employment (IS8)	174	7.191	25.233	−30.205	311.860
Forced labor (IS9)	174	6.840	10.691	−9.906	49.610
Freedom of association and collective bargaining (IS10)	174	5.390	7.537	−7.218	38.436
Human rights abuses, corporate complicity (IS11)	174	31.119	35.768	−43.287	366.427
Impacts on communities (IS12)	174	38.909	27.613	−9.005	130.754
Local participation issues (IS13)	174	5.864	11.357	−15.268	68.506
Occupational health and safety issues (IS14)	174	14.499	15.360	−8.114	92.968
Poor employment conditions (IS15)	174	19.961	29.630	−36.168	341.225
Social discrimination (IS16)	174	3.709	6.648	−5.995	42.380
Anti competitive practices (IG17)	174	14.116	13.848	−5.436	64.723
Corruption, bribery, extortion, money laundering (IG18)	174	15.386	17.651	−12.039	110.053
Executive compensation issues (IG19)	174	3.980	13.046	−19.018	104.190
Fraud (IG20)	174	22.985	19.869	−5.719	103.457
Misleading communication (IG21)	174	12.995	25.219	−33.468	259.695
Tax evasion (IG22)	174	3.588	20.991	−32.239	257.181
Tax optimization (IG23)	174	7.225	45.086	−52.528	336.364
Controversial products and services (IC24)	174	22.130	24.993	−18.292	149.561
Products (health environmental issues) (IC25)	174	24.189	22.637	−7.572	103.116
Supply chain issues (IC26)	174	28.198	23.500	−9.502	103.332
Violation of international standards (IC27)	174	4.808	11.836	−12.808	56.144
Violation of national legislation (IC28)	174	69.315	51.863	−6.903	382.872
Other ESG issues (IO29)	174	0.139	1.186	−2.259	10.508

This table presents summary statistics for individual indicators in the issue-based category. Each indicator is constructed according to Eq. (1). The 'Obs' indicates the number of months in our time-series analysis. The sample spans July 2007 to December 2021.

et al. (2020).[7] In addition, we account for various uncertainty variables, which include macroeconomic, financial, and real uncertainty as defined by Jurado et al. (2015),[8] as well as the Economic Policy Uncertainty (EPU) introduced by Baker et al. (2016)[9] and the Volatility Index (VIX).[10] Table A.2 in the appendix lists those predictors and provides corresponding brief descriptions.

3. Full-sample predictive analysis

In this section, we evaluate the full-sample (also known as in-sample) performance of the issue-based ESG market indicator on predicting excess market returns. Rapach and Zhou (2022) recommend using fewer independent variables in predictive regressions due to concerns about low efficiency and overfitting, both of which can arise when a large set of independent variables is incorporated into a single regression.

Accordingly, we first employ the single-predictor time-series regression using monthly data. The single-predictor predictive regression is as follows:

$$R_{t+1} = \theta_0 + \theta_1 X_t + \mu_{t+1} \quad for \quad t = 1, 2, \ldots T-1 \qquad (2)$$

where, R_{t+1} is the excess return on the S&P 500 index, X_t is one of variables described in Section 2.4 and Table A.2 in the appendix or ESG market indicator (ESG) at month t, and μ_{t+1} is the error term. The null hypothesis is that X does not have predictability ($\theta_1 = 0$) and the alternative hypothesis is that X can predict future returns ($\theta_1 \neq 0$).

The results of the single-predictor regression are reported in Table 2 under the panel labeled "Single-predictor Regressions". The coefficients associated with each variable are presented, along with the corresponding Newey–West t values and the adjusted R^2. The ESG market indicator exhibits a significant ability to forecast excess market returns. The coefficient of the ESG market indicator is 1.327 at the 1% significance level, and the adjusted R^2 is 7.902%.[11] We also test the in-sample forecasting performance of 28 ESG individual indicators and find that the majority of these individual indicators are not statistically significant.[12] This might be because, although the ESG individual indicators include information pertinent to future returns, the data may be rife with noise, hindering

[7] Unlike Atanasov et al. (2020), which use quarterly data, we use monthly data here. The consumption cycle is given by $c_t = \alpha + \beta_1 c_{t-72} + \beta_2 c_{t-73} + \beta_3 c_{t-74} + \beta_4 c_{t-75} + \epsilon_t$, where c is the U.S. monthly real expenditure per capita, downloaded from https://www.bea.gov/data/consumer-spending/main. See Atanasov et al. (2020) for further details.

[8] The data are available from https://www.sydneyludvigson.com/macro-and-financial-uncertainty-indexes.

[9] The data are available from https://www.policyuncertainty.com/.

[10] The data are available from https://www.cboe.com/tradable_products/vix/vix_historical_data/.

[11] The positive coefficient of the ESG market indicator ESG_t can be further explained by Tables B.3 and C.5 in the appendix. Generally, the values of γ are of the same sign as the coefficients of the individual indicators $esg_{i,t}$, leading predominantly to positive products. This results in ESG_t retaining a positive coefficient due to its positive correlation with the aggregate of these products.

[12] The results are shown in Table B.3 in the appendix.

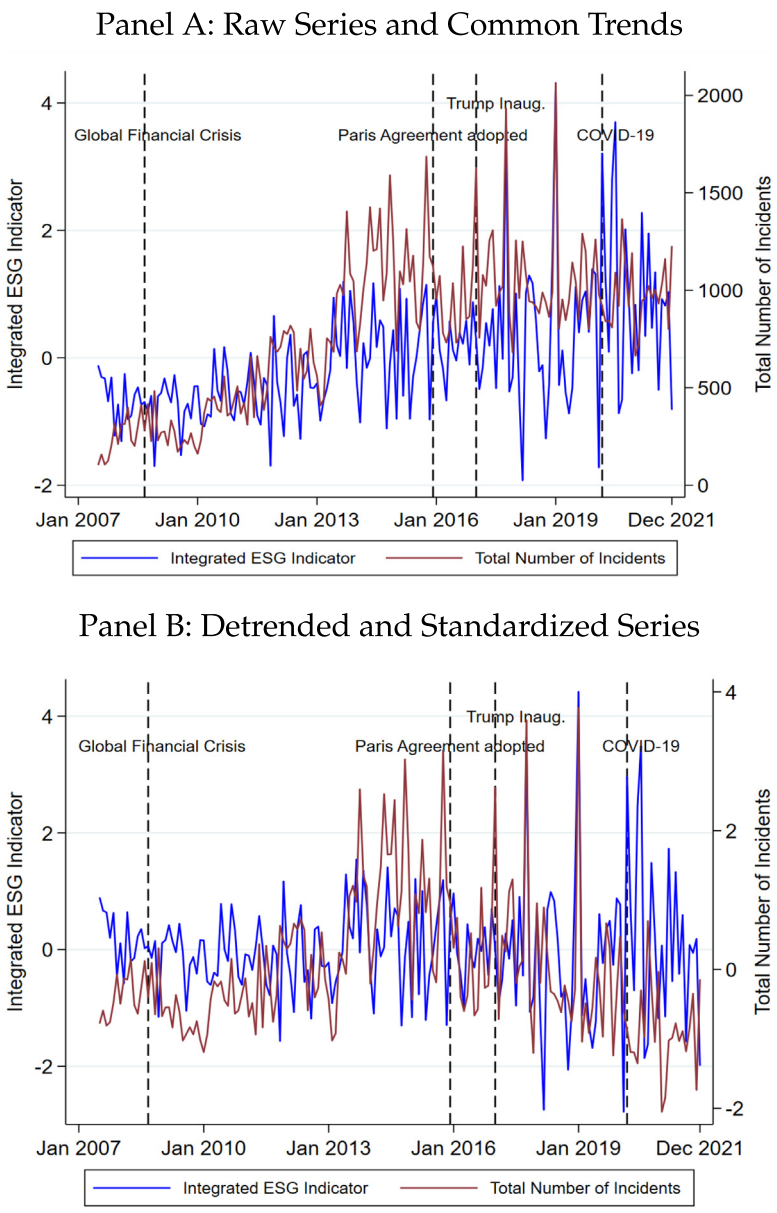


Fig. 1. Construction and Context of the ESG Predictive Indicator.
This figure plots the time series of our PLS-based ESG indicator and the simple sum of all incidents. Panel A shows the raw, non-detrended series, highlighting their shared low-frequency trend (correlation = 0.535). Panel B shows the corresponding detrended and standardized series, illustrating the much lower correlation (0.272) of the components used for month-to-month prediction. The series are annotated with major economic and political events. The sample period is from July 2007 to December 2021.

accurate prediction. The PLS method serves as a practical solution by extracting the most relevant information from the data. Regarding other predictive variables, only 4 of 20 variables have the power to predict returns.

To verify the robustness of the ESG market indicator's predictability, and to study whether the predictive power of the indicator stems from other fundamental variables (Rapach et al. 2010, Wang et al. 2019), we carry out two-predictor predictive regressions, controlling for other fundamental variables. The model is as follows:

$$R_{t+1} = \chi_0 + \chi_1 ESG_t + \chi_2 C_t + \psi_{t+1} \quad for \quad t = 1, 2, \ldots T-1 \tag{3}$$

where, ESG_t is the ESG market indicator at month t, C is the one of variables described in Section 2.4 and Table A.2 in the appendix.

Table 2 provides the results from the two-predictor regression analyses. The coefficients of the ESG market

Table 2
In-sample predictive regression of ESG market indicator.

	Single-predictor Regressions			Two-predictor Regressions				
Variables	θ_1	t	$Adj-R^2(\%)$	χ_1	t_1	χ_2	t_2	$Adj-R^2(\%)$
ESG	1.280***	3.597	7.400					
LNDE	0.335	0.338	−0.400	1.280***	3.593	0.329	0.348	6.900
LTY	−89.300**	−2.323	3.900	1.250***	3.568	−85.500**	−2.371	10.900
LNDY	0.736	0.227	−0.500	1.278***	3.580	0.431	0.139	6.800
LNDP	−0.023	−0.007	−0.600	1.281***	3.584	−0.211	−0.066	6.800
TBL	−78.500**	−2.180	2.300	1.286***	3.692	−79.400**	−2.467	9.700
LNEP	−0.490	−0.470	−0.400	1.282***	3.591	−0.509	−0.515	7.000
BM	1.728	0.283	−0.500	1.280***	3.595	1.624	0.280	6.900
INFL	100.000	1.112	0.100	1.287***	3.575	106.000	1.242	7.600
DFY	−60.100	−0.487	−0.200	1.283***	3.568	−62.700	−0.536	7.300
NTIS	41.062	1.589	2.200	1.272***	3.526	40.288	1.585	9.500
TMS	−17.900	−0.559	−0.300	1.274***	3.549	−14.600	−0.479	7.000
LTR	7.670	0.627	−0.300	1.284***	3.630	8.164	0.683	7.200
DFR	19.915	0.722	0.500	1.274***	3.396	19.141	0.740	7.800
SVAR	−14.100	−0.138	−0.500	1.323***	3.729	−37.100	−0.435	7.200
TRU	1.444	0.539	−0.300	1.271***	3.540	0.430	0.189	6.800
TFU	−2.589	−0.864	0.700	1.297***	3.556	−2.784	−0.991	8.300
TMU	−1.733	−0.455	−0.200	1.320***	3.587	−2.554	−0.749	7.600
EPU	0.016**	2.116	2.200	1.212***	3.546	0.013**	2.057	8.700
VIX	0.056	0.972	0.600	1.248***	3.593	0.044	0.860	7.600
C72	−12.600**	−2.433	2.900	1.269***	3.686	−12.400**	−2.396	10.200

This table reports the results of the single-predictor and two-predictor predictive regressions as follows,

$$R_{t+1} = \theta_0 + \theta_1 ESG_t + \mu_{t+1}$$

$$R_{t+1} = \chi_0 + \chi_1 ESG_t + \chi_2 C_t + \psi_{t+1} \quad for \quad t = 1, 2, \ldots T-1$$

where R_{t+1} is the future value of the excess return of the S&P 500 index and ESG_t is the ESG market indicator. C_t is one of the fundamental variables delineated in Section 2.4 and Table A.2 in the appendix. The data range from July 2007 to December 2021. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

indicator remain consistent, with values fluctuating between 1.23 and 1.39 across all regression models. These coefficients are statistically significant at the 1% level. Regarding other predictive variables, those that were significant in the single-predictor regression analysis continue to exhibit substantial predictive power for future excess returns. These variables include the long-term yield, treasury bill rate, economic policy uncertainty, and the consumption cycle. The in-sample R^2 values for the two-predictor regressions are notably higher than those for the corresponding single-predictor regressions. This suggests that the ESG market indicator provides both distinct and supplementary predictive information when compared to other well-established variables over the full sample period (Rapach et al., 2010).

4. Out-of-sample predictive analysis

Relying solely on in-sample analysis may not be sufficient to validate stock return predictability, as robust in-sample performance does not necessarily translate into successful out-of-sample forecasting. This disconnect can be attributed to factors such as estimation errors, parameter uncertainty, learning-based models, structural changes, etc. (e.g., Cochrane 2008, among others).

Welch and Goyal (2008) further note that, monthly, many macroeconomic predictors are not stationary, possibly stemming from over-fitting. This leads to the disappearance of their predictability in an out-of-sample context. Moreover, in a practical context, out-of-sample forecasting is more valuable, as it helps financial professionals make informed future investment decisions by providing insights into potential returns. In this section, we will use out-of-sample tests to explore whether real-time data of ESG incidents can offer tangible benefits to investors.

We use the recursive window technique to determine the coefficients and forecasts for the out-of-sample analysis, aligning with Chu et al. (2022), among others. Specifically, our process begins with an initial training period spanning 120 months and produces the first out-of-sample forecast. We then extend the initial training period by one month to generate the second forecast value.

To ensure that our out-of-sample (OOS) exercise constitutes a genuine forecasting procedure, free from lookahead bias, the entire process is performed recursively using an expanding window. For each forecast of return R_{t+1} made at the end of month t, all components of the model are estimated using only data available up to that point. This includes the full re-estimation of the PLS model to construct the latent ESG factor. The resulting factor at each step is then used directly in the predictive regression without any additional standardization or detrending. This rigorous, step-by-step re-estimation ensures that no information beyond time t, including future returns, future ESG values, or full-sample parameter estimates, is used in generating the forecast for R_{t+1}. This design faithfully mimics a real-time forecasting environment and ensures the integrity of our OOS results.

4.1. Statistical predictability

In accordance with prior studies (Jiang et al. 2019, Wang et al. 2018, among others), we employ the out-of-sample R^2 statistic (R^2_{OS} thereafter) introduced by Campbell and Thompson (2008), to evaluate out-of-sample predictive accuracy. Specifically, the R^2_{OS} statistic measures the percentage reduction in mean squared forecast error (*MSFE*) of a predictive model in comparison to a benchmark model. The formula of R^2_{OS} is given by:

$$R^2_{OS} = 1 - \frac{MSFE_{Predictive}}{MSFE_{Benchmark}} \tag{4}$$

$MSFE_{Predictive}$ and $MSFE_{Benchmark}$ are the mean squared forecast errors of a predictive model and a benchmark model, which are separately formulated as

$$MSFE_{Predictive} = \frac{1}{T-i}\sum_{t=i+1}^{T}(R_t - \hat{R}_t)^2 \tag{5}$$

$$MSFE_{Benchmark} = \frac{1}{T-i}\sum_{t=i+1}^{T}(R_t - \bar{R}_t)^2 \tag{6}$$

where R_t represents the realized excess market returns, $\hat{R}_t$ is the predicted return, $\bar{R}_t$ stands for the predicted return using a benchmark model, and i is the number of months in the initial training period, which in this paper is set at 120. A common benchmark employed is the historical mean of excess returns, expressed as $\bar{R}_t = \frac{1}{t-1}\sum_{q=1}^{t-1} R_q$. Intuitively, $R^2_{OS} > 0$ demonstrates that the predictive model's *MSFE* is less than that of the benchmark model, illustrating the superior performance of the former over the latter. This provides evidence of return predictability.

To further investigate whether the ability to predict returns leads to a statistically significant enhancement in terms of MSFE, we implement the Clark and West (2007) (*MSFE − adjusted*) test. Under this approach, the null hypothesis is that the MSFE from the benchmark forecasts is not greater than the MSFE stemming from the predictive model, formally expressed as $H_0 : R^2_{OS} \leq 0$.

We investigate the out-of-sample performance of our ESG market indicator, with the results reported in Table 3. We can find that the single-predictor predictive model incorporating the ESG market indicator yields an out-of-sample R^2_{OS} of 14.42%. This indicator is statistically significant at the 1% level, reflecting its robust ability to forecast market returns. In contrast to the majority of ESG individual indicators, which display limited predictive power in out-of-sample tests, the ESG market indicator markedly improves predictive performance.[13] For comparative analysis, we also present the performance of other variables. In the single-predictor model, only two out of 20 variables (EPU and VIX) demonstrate return predictability at the 5% level. When contrasted with the in-sample single-predictor regression, those indicators that are significant within the in-sample model lose their predictive ability in the out-of-sample context, consistent with Welch and Goyal (2008).

The performance of the combination of the ESG market indicator with a single control variable is also reported in Table 3. It is noteworthy that two-predictor models combining various popular variables with the ESG market indicator significantly outperformed the historical average benchmark in 17 of 20 cases, indicating pervasive return predictability. Besides, it should be emphasized that the two-predictor model does not consistently outperform the single-predictor models in an out-of-sample context. The inclusion of unrelated variables might hamper out-of-sample forecasting performance due to estimation errors, and an expansion in the parameter count might lead to fluctuating forecasts, thereby elevating forecasting variance (e.g., Rapach et al. 2010, Wang et al. 2019, Welch and Goyal 2008).

We also use ESG ratings provided by RepRisk to evaluate the predictive power of ESG factors. Interestingly, we find that the combined ESG market indicator is not significant in either single-predictor or two-predictor regressions. This result is inconsistent with the findings of Chu et al. (2022). These findings further support our arguments that different methodologies employed by various rating agencies can lead to non-robust results in the analysis of ESG return predictability.[14]

To summarize, consistent with the in-sample findings, the out-of-sample predictive ability of the ESG market indicator is statistically significant at the 1% level, regardless of whether single- or two-predictor tests are used. This suggests that our ESG market indicator encompasses vital information for predicting excess return.

To provide a more detailed analysis of our predictor's performance over time, we follow Welch and Goyal (2008) and plot the cumulative relative out-of-sample (OOS) performance in Fig. 2. The vertical axis shows the cumulative difference between the mean-squared forecast error (MSFE) of the historical mean benchmark and that of our ESG-based predictor. An upward-sloping line indicates periods when our ESG model outperforms the benchmark, while a downward-sloping line indicates periods of underperformance. This visualization not only clarifies whether our model adds predictive value but also helps identify when the predictor is most effective.

The time-varying performance shown in Fig. 2 is both economically intuitive and consistent with the forecasting literature, which highlights that predictive relationships naturally fluctuate across different economic regimes, regulatory environments, and shifts in investor sentiment. In particular, the relatively modest performance during the pre-2020 period coincides with a reduced emphasis on ESG policy under the Trump administration, which initiated the withdrawal from the Paris Agreement in 2017, with the withdrawal officially taking effect in November 2020 (Bolton & Kacperczyk, 2021). This diminished policy focus likely contributed to the low salience of ESG-related risks in the market, limiting investor responsiveness to ESG incident information during this time.

A significant structural break in predictive power emerges around February 2020, driven by two concurrent developments. First, the onset of the COVID-19 pandemic

[13] The results are reported in Table B.4 in the appendix.

[14] The results are available upon request.

Table 3
Out-of-sample predictive regression of ESG market indicator.

	Single-predictor Regressions		Two-predictor Regressions	
	R^2_{OS}	*MSFE − adjusted*	R^2_{OS}	*MSFE − adjusted*
ESG	14.420***	2.651		
LNDE	0.279	0.881	14.424***	2.589
LTY	4.099*	1.504	14.229***	2.496
LNDY	−2.550	−0.732	12.858**	2.341
LNDP	−0.971	−0.622	14.245***	2.461
TBL	1.404	1.042	15.245***	2.612
LNEP	0.457	1.159	14.060***	2.597
BM	−2.180	−0.815	13.895***	2.450
INFL	−1.972	−0.641	10.800**	2.287
DFY	−0.603	−0.269	13.891***	2.624
NTIS	1.557*	1.351	14.166***	2.816
TMS	−1.144	−0.802	12.501***	2.441
LTR	−1.432	−0.278	13.038**	2.316
DFR	−2.413	−0.230	11.840***	2.458
SVAR	−32.376	−1.028	−10.556	0.169
TRU	−26.670	−0.482	−8.022	0.423
TFU	−6.588	−0.432	10.824**	2.349
TMU	−17.977	−0.783	3.802	1.165
EPU	4.573**	1.702	13.921***	2.554
VIX	2.789**	1.837	16.709***	2.597
C72	−1.345	0.354	14.422**	2.180

This table reports out-of-sample performance for single-predictor and two-predictor predictive regressions. ESG is the ESG market indicator. The other fundamental variables are delineated in Section 2.4 and Table A.2 in the appendix. When forecasting returns for period $t+1$, the data is confined to the information available up to period t. The single-predictor regression is the model with only one predictor, and the two-predictor regression is the model combining the ESG market indicator with one of the remaining predictors. R^2_{OS}, introduced by Campbell and Thompson (2008), is the percentage reduction in mean squared forecast error of the predictive regression in comparison to the benchmark model, calculated according to Eq. (4). *MSFE − adjusted*, introduced by Clark and West (2007), is the statistics for testing hypothesis $H_0 : R^2_{OS} \leq 0$ against $H_A : R^2_{OS} > 0$. The initial training period for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

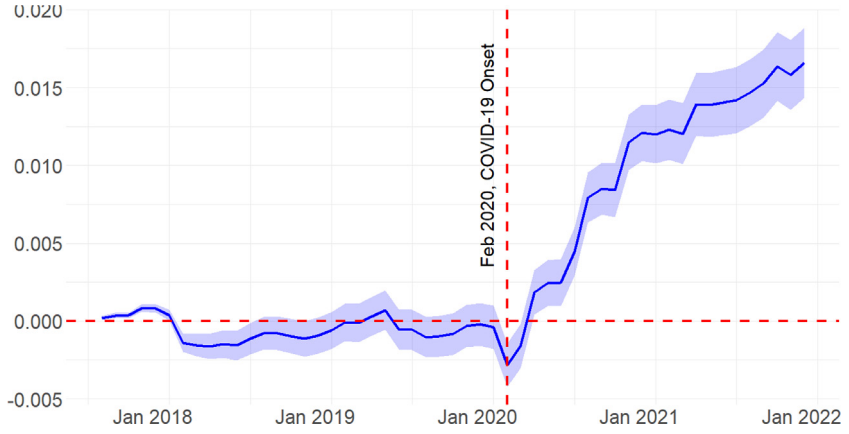


Fig. 2. Cumulative Relative Out-of-Sample Performance.
This figure plots the cumulative sum of the difference between the squared forecast errors (MSFEs) for the historical mean benchmark and our ESG predictive model, following Welch and Goyal (2008). An upward-sloping line indicates periods during which the ESG model outperforms the benchmark. The out-of-sample period shown is from July 2017 to December 2021.

drastically raised the salience of ESG-related physical risks (e.g., supply chain vulnerabilities and employee welfare), transforming ESG considerations from firm-specific concerns into systemic risk factors (Cheema-Fox et al., 2021). Second, the political shift in early 2021 with the inauguration of the Biden administration reintroduced substantial transition risks through renewed climate and ESG-related regulatory commitments, including rejoining the Paris Agreement (Pástor et al., 2022). This new economic environment created conditions for prolonged market underreaction, a well-documented channel of predictability (Glossner, 2021; Hong & Stein, 1999).

Rather than indicating inconsistency or unreliability, the time-varying nature of our predictor's performance demonstrates its responsiveness to changing macroeconomic and regulatory conditions. This strengthens both the theoretical foundation and the practical relevance of our forecasting model.

Finally, to assess the persistence of this predictability, we also test the performance of our indicator for longer

forecast horizons (3, 6, and 12 months).[15] We find that its predictive power is concentrated at the one-month horizon, with the longer-horizon results proving statistically insignificant based on the Clark–West test. This suggests that the information captured by our incident-based signal is relatively short-lived and is quickly incorporated into market prices. This characteristic distinguishes it from slower-moving valuation-based predictors.

4.2. Economic predictability

In this section, we examine the economic performance associated with stock market return forecasts generated by the ESG market indicator, with a specific focus on asset allocation strategies. In addition to evaluating forecast accuracy, it is also crucial to explore whether predictions derived from the ESG market indicator can enhance an investor's trading performance. Although there exists a generally positive correlation between forecast accuracy and trading performance, they do not invariably correspond perfectly (Campbell & Thompson, 2008). Following Campbell and Thompson (2008), Gao et al. (2018), Jiang et al. (2019) and Chu et al. (2022), among others, we employ two methods to investigate economic performance: the certainty equivalent return (CER) gain and the Sharpe ratio. We model a mean–variance investor who is risk-averse and allocates her wealth between risky assets and risk-free Treasury bills to optimize her utility.

The investor's portfolio return for month $t+1$ is formulated as:

$$R_{p,t+1} = \omega_t R_{t+1} + (1-\omega_t) r_{f,t+1} \tag{7}$$

where, $R_{p,t+1}$ is the investor's portfolio returns for month $t+1$; R_{t+1} represents the risky assets' returns for the same month; $r_{f,t+1}$ denotes the risk-free rate for that period; and ω_t is the weight allocated to risky assets in the portfolio.

Monthly, the investor adjusts the weight of risky assets (ω_t) to maximize expected utility. This adjustment is based on forecasts of the return on risky assets for the subsequent month. The optimal weight, ω_t^*, at the end of month t is determined by:

$$\omega_t^* = \frac{1}{\gamma}\frac{\hat{R}_{t+1}}{\hat{\sigma}_{t+1}^2} \tag{8}$$

where γ refers to the investor's risk aversion, $\hat{R}_{t+1}$ and $\hat{\sigma}_{t+1}^2$ are the forecasts for excess market returns and their variance, respectively. We set $\gamma = 3$.[16]

Variance forecasts are calculated using a five-year rolling window of historical returns. Additionally, the optimal weight ω_t is constrained to lie within the range of -1 to 2. This constraint allows the investor to either short sell up to 100% or leverage their position by 100% through margin borrowing.[17]

The annualized CER percentage over the entire out-of-sample forecasting period is expressed as:

$$CER = 1200\bar{R}_p - 0.5\gamma\hat{\sigma}_p^2 \tag{9}$$

where $\bar{R}_p$ and $\hat{\sigma}_p^2$ are the mean and variance, respectively, of the realized portfolio returns over the out-of-sample period. The CER can be interpreted as the investor's compensation for engaging in the stock market. The CER gain, the difference between the CER values of the predictive model and a benchmark model, measures the economic significance of the predictive model. In addition to CER gains, trading performance is further evaluated using the Sharpe ratio. This is calculated as the portfolio's annualized average excess return, normalized by its standard deviation.

The economic performance derived from our ESG market indicator is reported in Table 4. In single-predictor regression, the ESG market indicator yields a CER gain of 13.87% and a Sharpe ratio of 1.306. These figures surpass the corresponding economic values for most other variables, with the notable exception of default return variance (DFR). This indicates that the ESG market indicator serves as an effective predictor of excess market returns.

In two-predictor models, all combinations yield positive CER gains and high Sharpe ratios over the full out-of-sample period. Compared with their single-predictor counterparts, the combination of the ESG market indicator with an additional variable often improves economic performance. The most substantial economic gains are achieved when the ESG market indicator is combined with the book-to-market ratio, yielding a CER gain of 19.358% and a Sharpe ratio of 1.603. Interestingly, variables that exhibit negative CER gains in single-predictor settings shift to positive gains when ESG information is incorporated. Except for the long-term yield model, the predictive power of most models improves through incorporating the ESG market indicator. In summary, our results strongly suggest that using the ESG market indicator to forecast market returns can offer investors substantial economic benefits compared to relying solely on traditional variables.

4.3. Predictability of ESG components

To further investigate which specific ESG components significantly influence excess market returns, this section breaks down the composite ESG variable into four distinct dimensions: Environmental (E), Social (S), Governance (G), and Cross-cutting (C). Each of these dimensions is constructed from issue-specific individual indicators, using the PLS technique.

Table 5 reports the out-of-sample forecasting performance of the four ESG components. Our results reveal that the social dimension emerges as the most robust predictor, both statistically and economically. Specifically, the predictive regression model that includes the social variable yields an out-of-sample R_{OS}^2 of 9.607%. This variable is statistically significant at the 5% level, highlighting its considerable predictive power for market returns. Additionally, this component generates a

[15] The results are available upon request.

[16] This is in line with Rapach et al. (2010) and Dangl and Halling (2012). In Section 6, we also consider other risk aversion coefficients, namely $\gamma = 5$ and 7.

[17] In Section 6, we impose more stringent constraints, limiting the allocation between 0 and 1, thereby prohibiting short selling and not allowing for financial leverage.

Table 4
Economic performance of ESG market indicator.

	Single-predictor Regressions		Two-predictor Regressions	
	CER Gain	Sharpe Ratio	CER Gain	Sharpe Ratio
ESG	13.870	1.306		
LNDE	0.083	0.771	14.521	1.346
LTY	5.820	0.969	3.123	0.884
LNDY	−2.728	0.678	18.261	1.561
LNDP	−2.101	0.682	16.165	1.448
TBL	6.806	1.081	19.299	1.553
LNEP	0.687	0.794	13.521	1.295
BM	−3.199	0.709	19.358	1.603
INFL	−1.103	0.724	10.099	1.153
DFY	−1.572	0.715	13.101	1.259
NTIS	3.018	0.892	16.621	1.455
TMS	−1.162	0.727	15.578	1.393
LTR	2.156	0.849	14.168	1.278
DFR	12.294	1.356	16.288	1.435
SVAR	−4.995	0.566	3.427	0.897
TRU	−1.079	0.724	10.893	1.209
TFU	−3.367	0.629	12.381	1.291
TMU	−3.397	0.624	12.295	1.248
EPU	2.723	0.868	11.392	1.202
VIX	4.733	0.939	14.153	1.327
C72	−6.482	0.479	13.855	1.340

This table reports the out-of-sample economic performance of the forecasts generated by single-predictor and two-predictor predictive regressions in Table 3. ESG is the ESG market indicator. The other fundamental variables are delineated in Section 2.4 and Table A.2 in the appendix. "CER Gain" is the difference of the certainty equivalent return (CER) between predictive models and the benchmark model. "Sharpe Ratio" is calculated as the annualized average excess return of a portfolio divided by its annualized standard deviation. See Section 4.2 for further details. The initial training window for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021.

Table 5
Out-of-sample predictive regression of ESG components.

	R^2_{OS}	*MSFE − adjusted*	CER Gain	Sharpe Ratio
Environmental	2.774**	2.112	4.974	0.967
Social	9.607**	2.097	13.288	1.276
Governance	1.200	0.878	6.966	1.052
Cross-cutting	2.145*	1.487	3.092	0.880

This table reports the out-of-sample forecasting performance of ESG components. These components are generated from the individual ESG predictors in corresponding categories. R^2_{OS}, introduced by Campbell and Thompson (2008), is the percentage reduction in mean squared forecast error of the predictive regression in comparison to the benchmark model, calculated according to Eq. (4). *MSFE − adjusted*, introduced by Clark and West (2007), is the statistics for testing hypothesis $H_0 : R^2_{OS} \leq 0$ against $H_A : R^2_{OS} > 0$. "CER Gain" is the difference of the certainty equivalent return (CER) between predictive models and the benchmark model. "Sharpe Ratio" is calculated as the annualized average excess return of a portfolio divided by its annualized standard deviation. See Section 4.2 for further details. The initial training period for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

CER gain of 13.288% and a Sharpe ratio of 1.276, figures that closely align with the economic values derived from the ESG market indicator. While the environmental and cross-cutting dimensions are statistically significant, they exhibit comparatively weaker predictability. Their respective out-of-sample R^2_{OS} values are 2.774% and 2.145%, and both dimensions achieve CER gain exceeding 3%. In summary, among the three components (E, S, and C), the social component contributes most significantly to the predictive performance of the ESG market indicator. Interestingly, the governance variable shows no substantial predictive ability for market returns.

4.4. Performance on portfolio returns

The concept that certain ESG incidents are more relevant to specific industries while being inconsequential for others is intrinsically logical (Henriksson et al., 2019). For example, CO_2 emissions may be highly significant for utility sectors but may have limited relevance for firms in the financial domain. Hong and Kacperczyk (2009) argue that sin stocks typically yield higher expected returns than other stock categories. It is noteworthy that these sin stocks predominantly originate from distinct sectors, such as the tobacco industry. This raises an important question: To what extent does the predictive performance of our ESG market indicator apply across diverse industry portfolios? Is its influence widespread, or does it mainly pertain to specific sectors? In the following section, we investigate the potentially differentiated predictive ability of the ESG market indicator across 49 industry categories,

as categorized and sourced from the Ken French data library.[18]

Panel A of Table 6 reveals that the predictive ability of our ESG indicator is remarkably widespread, showing a statistically significant positive relationship with future returns for 47 of the 49 industry portfolios. This suggests that the information captured by our signal represents a systematic, market-wide risk factor rather than being confined to a few specific sectors. The magnitude of the estimated coefficients (β) varies economically intuitively. For instance, the indicator has a particularly strong effect on industries with complex global supply chains, high consumer visibility, and significant operational leverage, such as Automobiles and Trucks ($\beta = 3.304$), Electrical Equipment ($\beta = 2.151$), and Business Services ($\beta = 1.737$). This is consistent with the view that such industries are more sensitive to the reputational and operational risks signaled by adverse ESG incidents. In contrast, the two exceptions, Precious Metals and Utilities, are sectors whose returns are often driven by unique factors such as commodity prices and regulatory frameworks, respectively, potentially overwhelming the ESG signal.

In addition, we extend our analysis to examine the ESG market indicator's ability to forecast returns for specific factor portfolios. We consider various factor portfolios including the size (SMB), value (HML), profitability (RMW), and investment (CMA) factor portfolios from Fama and French (2015) five-factor model (FF5 therefore), the momentum (MOM) factor portfolio from Carhart (1997),[19] and the size (ME), investment (I/A), return on equity (ROE), and expected growth (EG) factor portfolios from Hou et al. (2015) q-factor model.[20]

Panel B of Table 6 examines the link between our ESG indicator and standard risk factors. The results provide further insight into the nature of the predictability. Notably, we find a significant negative relationship between our ESG indicator and the investment factors CMA ($\beta = -0.310$) and I/A ($\beta = -0.371$). This suggests that a higher level of our ESG signal, indicating a heightened risk environment, predicts lower future returns for firms that invest conservatively (the "investment" factor). This is consistent with asset-pricing theories, which hold that perceptions of risk directly influence corporate investment decisions and subsequent returns. The lack of a significant relationship with most other factors, such as size (SMB) and value (HML), indicates that our ESG predictive signal captures a distinct source of risk not spanned by these classic factors.

5. Characterizing the ESG predictive signal

Our primary analysis establishes that an indicator constructed from 28 ESG incident streams using Partial Least Squares (PLS) has significant predictive power for aggregate market returns. We now turn to investigating the economic channels that might drive this predictability.

A key feature of the PLS methodology is that it constructs a latent factor for maximum predictive covariance, rather than for direct structural interpretation. While this makes it a powerful forecasting tool, the resulting ESG predictive indicator does not have a simple, one-to-one mapping to a single economic concept, such as "aggregate ESG risk" or "ESG sentiment". It is, by construction, a statistical composite.

This is not merely a theoretical point; it is confirmed empirically. A crucial finding from our analysis (detailed in Section 2.3) is that our PLS-based indicator has only a weak correlation (0.272) with the simple, detrended sum of all reported incidents. This striking result implies that a naive approach of simply aggregating all adverse news fails to capture the predictive signal, as the raw news stream is too noisy. The PLS method, therefore, acts as an essential economic filter, identifying the specific, nuanced combination of the 28 incident types that is most informative about future returns.

Given that our indicator is a statistically optimized signal rather than a direct measure of an economic state, our approach in this section is to characterize its predictive properties. We ask: what is the economic nature of the information that this 'smart filter' isolates? Does it relate to future cash flows, discount rates, market risk, or firm fundamentals? By answering these questions, we can diagnose the channels through which our indicator's predictive power operates. The following subsections explore these distinct economic characteristics.

5.1. The signal's link to cash flow and discount rate news

A fundamental principle in asset pricing is that any variable with predictive power for stock returns must do so by providing information about either future cash flows (the "cash flow channel") or future discount rates (the "discount rate channel"), or both (e.g., Cochrane 2011). Decomposing the returns our signal predicts into these two components is therefore a powerful first step in characterizing its economic nature.

Both channels are theoretically plausible. The cash flow channel is supported by the literature, which shows that material ESG events are tied to future profitability and real costs (e.g., Derrien et al. 2025, Khan et al. 2016). The discount rate channel is supported by theories linking ESG to systemic risk perceptions (e.g. Albuquerque et al. 2019). By testing the link between our PLS indicator and each news component, we can determine whether our signal primarily reflects future corporate profitability or changes in market risk premia.

To perform the return decomposition, we employ a first-order Vector Autoregression (VAR) framework developed by Campbell (1991), Campbell and Vuolteenaho (2004).[21] Chen and Zhao (2009) have demonstrated that this decomposition is highly sensitive to the choice of predictors. To address this concern, we adopt strategies

[18] The data are available from http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

[19] The data are available from http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

[20] The data are available from http://global-q.org/factors.html.

[21] See Appendix D to find the details.

Table 6
Portfolio return predictability.

Panel A: Industry Portfolios							
Industry	β	t	$adj-R^2(\%)$	Industry	β	t	$adj-R^2(\%)$
Agriculture	0.983*	1.847	1.356	Machinery	1.491***	3.199	3.735
Aircraft	2.003***	2.947	7.460	Measuring and Control Equipment	1.300***	3.356	4.976
Apparel	1.450***	3.115	4.340	Medical Equipment	1.126***	2.986	4.115
Automobiles and Trucks	3.304***	2.847	9.576	Non-Metallic and Industrial Metal Mining	1.804***	2.937	2.959
Banking	1.434***	2.663	3.551	Personal Services	1.239**	2.301	3.124
Beer & Liquor	0.813**	2.395	3.519	Petroleum and Natural Gas	1.695*	1.671	4.243
Business Services	1.737***	3.552	9.625	Pharmaceutical Products	0.838***	2.892	3.363
Business Supplies	1.203***	3.462	4.372	Precious Metals	0.447	0.573	−0.415
Candy & Soda	0.766*	1.795	1.128	Printing and Publishing	1.659***	3.179	4.828
Chemicals	1.653***	3.217	5.632	Real Estate	1.736**	2.394	2.574
Coal	2.154**	2.099	1.818	Recreation	1.964***	3.066	6.026
Communication	1.163***	3.251	5.370	Restaurants, Hotels, Motels	1.176**	2.036	5.441
Computers Hardware	1.047**	2.466	2.013	Retail	1.206***	2.783	5.970
Computers Software	1.467***	3.567	7.212	Rubber and Plastic Products	1.428***	3.179	4.328
Construction Materials	1.970***	3.655	6.037	Shipbuilding, Railroad Equipment	1.603***	2.899	2.951
Construction	1.922**	2.545	6.010	Shipping Containers	0.833**	2.132	1.525
Consumer Goods	0.866***	2.807	3.840	Steel Works Etc	1.707***	2.657	2.715
Defense	1.372***	3.980	5.161	Textiles	2.078**	2.602	3.313
Electrical Equipment	2.151***	4.053	8.846	Tobacco Products	1.169***	2.759	4.028
Electronic Equipment	2.026***	4.331	9.506	Trading	1.725***	3.854	5.822
Entertainment	1.998***	3.841	4.676	Transportation	1.683***	4.074	7.615
Fabricated Products	1.876**	2.538	3.719	Utilities	0.525*	1.693	1.136
Food Products	0.661**	2.006	2.321	Wholesale	1.430***	3.473	6.743
Healthcare	1.454***	2.756	5.065	Other	0.942**	2.278	1.820
Panel B: Factor Portfolios							
Factor	β	t	$adj-R^2(\%)$	Factor	β	t	$adj-R^2(\%)$
SMB	0.307	1.345	0.008	ME	0.311	1.359	0.008
HML	−0.168	−0.591	−0.003	I/A	−0.371***	−2.807	0.035
RMW	0.185	1.256	0.004	ROE	−0.287	−1.455	0.005
CMA	−0.310***	−2.906	0.030	EG	−0.195	−0.756	0.001
MOM	−0.283	−0.897	−0.002				

This table reports the results of the single-predictor predictive regression of the ESG market indicator on industry portfolio returns and risk factor portfolio returns. The future values of the industry or factor portfolio's excess returns are dependent variables, and the ESG market indicator is the independent variable. The 49 industry portfolios are defined and sourced from the Ken French data library. The factors considered in the table are the size (SMB), value (HML), profitability (RMW), and investment (CMA) factor portfolios from (Fama & French, 2015) five factor model (FF5 therefore), the momentum (MOM) factor portfolios from (Carhart, 1997) and the corresponding size (ME), investment (I/A), return on equity (ROE), expected growth (EG) factor portfolios from (Hou et al., 2015) q-factor model—the data range from July 2007 to December 2021. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

from the approaches of Rapach et al. (2016) and Chu et al. (2022) to broaden the set of variables for the VAR model.

Upon obtaining these component variables, we employ predictive regression analyses to investigate the mechanisms through which the ESG market indicator forecasts stock market returns (Cochrane, 2011). The predictive regressions for cash flow news and discount rate news are formally specified as follows:

$$\begin{aligned} CFN_{t+1} &= \beta_{CFN} ESG_t + \mu_{CFN,t+1} \\ DRN_{t+1} &= \beta_{DRN} ESG_t + \mu_{DRN,t+1} \end{aligned} \tag{10}$$

By integrating Eqs. (D.8) and (10), it is apparent that the estimated coefficient $\hat{\beta}$ of the ESG market indicator in the baseline regression, represented as Eq. (C.1), is given by $\hat{\beta}_E + \hat{\beta}_{CFN} - \hat{\beta}_{DRN}$. Here, $\hat{\beta}_E$ is estimated from the regression $E_t(R_{t+1}) = \alpha_E + \beta_E ESG_t + \mu_{E,t+1}$. Therefore, by estimating the parameters in Eq. (10), we can understand which specific channels enable the ESG market indicator to predict future excess market returns.

Table 7 presents the estimated coefficients and corresponding t-statistics from the predictive regressions in Eq. (10). The results reveal a key characteristic of our predictive signal: it contains significant forward-looking information about both future cash flows and discount rates. Market returns are decomposed using various VAR model specifications. All estimated coefficients $\hat{\beta}_{CFN}$ are statistically significant at the 5% level or better and account for the majority of the overall predictive effect (with $\hat{\beta}$ reported as 1.280 in Table 2). In contrast, the $\hat{\beta}_{DRN}$ estimates are generally smaller in magnitude and constitute a minor portion of $\hat{\beta}$ in most cases. Moreover, out of 17 model specifications, only 10 yield statistically significant $\hat{\beta}_{DRN}$ estimates.

Taken together, these results indicate that the forecasting ability of our ESG market indicator operates through both the cash flow and discount rate channels but is predominantly driven by the cash flow channel. This broader pattern is consistent with the nature of our incident-based signal, which is constructed from discrete real-world events that are likely to result in tangible future costs. For example, a negative ESG incident may lead to higher future liabilities, both from addressing the immediate incident and from increased expectations of similar incident-related costs. It may also trigger greater regulatory scrutiny and compliance costs, or reputational

Table 7
Cash flow and discount rate channels.

Variable Y	$\hat{\beta}_{CFN}$	t-statistics	$\hat{\beta}_{DRN}$	t-statistics
R+CV(2004)	1.376***	3.538	0.104	1.285
R+CZ(2009)	1.305***	3.499	−0.017	−0.122
R+LNDP	1.249***	3.717	−0.065***	−2.667
R+LNDP+ LNDY	1.145***	3.218	−0.158***	−3.060
R+LNDP+ EP	1.224***	3.748	−0.092***	−2.747
R+ LNDP+ LNDE	1.224***	3.748	−0.092***	−2.747
R+ LNDP+ RVOL	1.254***	3.723	−0.057**	−2.488
R+ LNDP+ BM	0.527**	2.498	−0.811***	−3.694
R+ LNDP+ NTIS	0.463**	2.253	−0.837***	−3.474
R+ LNDP+ TBL	1.110***	4.057	−0.203	−1.289
R+ LNDP+ LTY	1.142***	3.652	−0.118	−0.423
R+ LNDP+ LTR	1.306***	3.806	−0.014	−0.490
R+ LNDP+ TMS	1.310***	3.581	0.010	0.122
R+ LNDP+ DFY	0.641***	3.145	−0.676***	−2.793
R+ LNDP+ DFR	1.232***	3.560	−0.084***	−2.946
R+ LNDP+ INFL	1.230***	3.957	−0.094	−1.424
R+ LNDP+ PCA	0.371**	2.088	−0.923***	−3.572

This table reports the results of predictive regression models as follows

$$CFN_{t+1} = \beta_{CFN} ESG_t + \mu_{CFN,t+1}$$

$$DRN_{t+1} = \beta_{DRN} ESG_t + \mu_{DRN,t+1}$$

where CFN_{t+1} and DRN_{t+1} are the future cash flow news and discount rate news, respectively. We follow the VAR framework as presented by Campbell (1991) and Campbell and Vuolteenaho (2004) to decompose the excess returns of the S&P 500 index into cash flow news (CFN) and discount rate news (DRN). Various VAR specifications are employed, and the variables used in each VAR model are indicated by the row names: R is the excess returns of the S&P 500 index. CV(2004) means that the variables identified by Campbell and Vuolteenaho (2004) are included in the VAR model. They are the bond yield spread, Shiller's price-to-earnings ratio for the S&P 500 index, and the small-stock value spread. CZ(2009) represents that the first five components from a range of macroeconomic variables used in Chen and Zhao (2009) are added in the VAR model. These variables include term spread, Shiller's price-to-earnings ratio, small-stock value spread, credit spread, long-term corporate bond return, book-to-market ratio, 3-month T-bill rate, inflation, corporate issuing activity, and stock variance. The following VAR model specifications include log market returns, log dividend-price ratio, and/or one of the remaining 13 macroeconomic variables as identified in Welch and Goyal (2008). The last test incorporates the first component extracted from these 13 macroeconomic variables into the VAR framework, alongside the log market returns and log dividend-price ratio. See Section 5.1 for further details. The data range from July 2007 to December 2021. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

damage resulting in lower sales (e.g., Derrien et al. 2025, Pedersen et al. 2021). Consequently, our indicator primarily captures forward-looking information about cash flows, providing a direct economic link between ESG incidents and market returns.

5.2. Characterizing the signal's performance across economic cycles

A key characteristic of any return predictor is how its performance varies with the state of the economy. Seminal works have shown that the predictive ability of many variables is time-varying and often stronger during economic downturns, when risk aversion is high (e.g., Campbell and Cochrane 1999, Rapach et al. 2010). Therefore, to further characterize our ESG predictive indicator, we investigate whether its forecasting power is uniform across the business cycle or exhibits a cyclical pattern. This analysis helps determine whether the information captured by our signal is related to time-varying risk premiums central to asset pricing.

To test for this state-dependency, we incorporate various measures of the economic cycle, along with their interactions with our ESG indicator, into the main predictive regression. This allows us to assess whether the forecasting relationship is amplified or dampened during recessions. More precisely, the regression equation is given by:

$$R_{t+1} = \alpha + \beta_{ESG} ESG_t + \beta_{EC} EC_t + \beta_{ESG\times EC} ESG_t \cdot EC_t + \epsilon_{t+1} \quad (11)$$

where EC_t is an indicator for the economic cycle. We consider five alternatives for defining recession periods. The first two are dummy variables set to 1 during recession periods and 0 otherwise, based on classifications by the National Bureau of Economic Research (NBER) and the Organisation for Economic Co-operation and Development (OECD). We also consider continuous variables for economic cycle indicators, such as the consumption cycle as used in Atanasov et al. (2020),[22] the Chicago Fed National Activity Index (CFNAI),[23] and the U.S. Recession Probabilities (USRPs).[24] The coefficient of interest is $\beta_{ESG\times EC}$, which captures the marginal predictive power of our ESG

[22] The data are available from https://www.bea.gov/data/consumer-spending/main.

[23] Data obtained from the Federal Reserve Bank of Chicago: https://www.chicagofed.org/research/data/cfnai.

[24] USRPs are derived from a dynamic-factor Markov-switching model in Chauvet and Piger (2008) using four monthly indicators: non-farm

indicator during different economic states. A significant coefficient would indicate that the signal's effectiveness is not constant, but instead tied to the business cycle.

Table 8 displays the results. The β_{ESG}s are statistically significant at the 5% level, irrespective of the various methodologies used to define a recession. This finding is consistent with insights from Table 2 and confirms the robust nature of the ESG market indicator as a predictor. In the regression that included the NBER recession variable, the interaction term is positively significant at the 1% level. The effective coefficient of ESG_t, calculated as $\beta_{ESG} + \beta_{ESG\times EC}$, equals 5.384 during a recession. This suggests a stronger forecasting ability of the ESG market indicator during economic downturns, as recognized by the NBER. Similar findings are observed when using U.S. recession probabilities to define economic cycles; a single-unit change in the ESG indicator increases excess market returns from 1.069% to 1.115% during an economic trough. These patterns are consistent with studies by Rapach et al. (2010) and Li and Tsiakas (2017). Furthermore, predictive regressions based on the NBER recession dummy and the U.S. recession probabilities, along with their interaction terms, yield the highest $adj - R^2$ values, explaining future market returns more effectively than all other regressions specified in Eqs. (2) and (3).

This finding reveals another key characteristic of our predictive signal: its forecasting power appears to be strongly counter-cyclical. However, it is important to note that this effect is not universal across all definitions of the economic cycle. The interaction terms for the other three cycle indicators are not statistically significant, indicating that the heightened predictability is most closely aligned with the widely used NBER and U.S. recession probability definitions. This suggests that the information captured by our indicator becomes particularly salient and impactful during the sharp, distinct periods of heightened market-wide risk and investor anxiety captured by these specific downturn measures, consistent with theories of time-varying risk premiums.

5.3. Characterizing the signal's link to market volatility

A key part of characterizing our ESG predictive indicator is to understand its relationship with market-wide risk and uncertainty. Literature reveals a significant relationship between stock returns and their associated volatility, or market volatility. The empirical findings in this area, however, are diverse. Some research works highlight consistently positive risk premium estimates (e.g., Ghysels et al. 2005). Conversely, other studies find negative correlations, fluctuating outcomes, or estimate that lack significance (e.g., Adrian and Rosenberg 2008, and Bollerslev et al. 2020). Recent literature has begun to link firm-level ESG characteristics to lower stock volatility (e.g. Liu et al. 2023). We therefore aim to diagnose our predictive signal by testing whether it contains forward-looking information about future changes in market volatility.

To test this hypothesis directly, we employ a 'level-on-change' regression framework. This model examines whether the current level of our ESG indicator predicts future changes in market volatility, providing a clear test of its role as a leading indicator of market risk dynamics. The model is specified as:

$$\Delta MV_{t+1} = \beta_0 + \beta_1 ESG_t + \beta_2 \Delta MV_t + \epsilon_{t+1} \tag{12}$$

Here, ΔMV_{t+1} is the one-month change in market volatility, defined as the change in $\log(\sqrt{SVAR})$, where $SVAR$ is defined in Table A.2 in the appendix. A significant coefficient for β_1 would provide direct evidence that our return predictor is also a powerful volatility predictor.

The results are presented in Panel A of Table 9. The ESG market indicator has a coefficient of -0.102, significant at the 5% level. This negative coefficient is economically meaningful: it indicates that a higher level of our ESG predictive indicator is a strong signal of a decline in market volatility the following month.

This finding provides a powerful economic underpinning for our main result of positive return predictability. Our analysis reveals a contemporaneous correlation of -0.472 between changes in market volatility and the excess return of the S&P 500 in our data set, consistent with the well-documented negative volatility risk premium (e.g., Bollerslev et al. 2020). Therefore, by forecasting a decline in future volatility, our ESG indicator is simultaneously forecasting the higher returns associated with such a calming of the market. This supports the idea that the ability to forecast market risk is a key channel through which our ESG indicator's predictive power operates.

5.4. Characterizing the signal's link to firm fundamentals

Beyond its link to market-wide risk perceptions, we now test whether our ESG predictive signal contains forward-looking information about the tangible, real economy by examining its ability to forecast future changes in aggregate firm fundamentals. A key characteristic of a meaningful economic predictor, as opposed to one driven by sentiment or temporary market frictions, is its connection to future corporate performance. Our central hypothesis is that the nuanced information captured by our signal serves as a leading indicator of future corporate performance. This aligns with theories such as Pedersen et al. (2021), who argue that ESG information is most relevant for asset prices when it is linked to fundamentals, and recent evidence from Derrien et al. (2025) connecting firm-level ESG news to future cash flows.

To test this hypothesis rigorously, we continue to employ the 'level-on-change' framework established in the previous section. This approach addresses the non-stationarity of the fundamental variables and tests whether the level of our ESG indicator (ESG_t) can predict future one-period changes (ΔFF_{t+1}) in a wide range of fundamentals grouped into logical categories: Profitability, Valuation, Financial Health, and Efficiency. The model is specified as:

$$\Delta FF_{t+1} = \beta_0 + \beta_1 ESG_t + \beta_2 \Delta FF_t + \epsilon_{t+1} \tag{13}$$

payroll, the industrial production index, real personal income excluding transfers, and real manufacturing/trade sales.

Data is sourced from the Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/RECPROUSM156N.

Table 8
Economic cycle and ESG.

	β_{ESG}	β_{EC}	$\beta_{ESG\times EC}$	$adj-R^2(\%)$
$NBER_{dummy}$	1.017***	−2.892*	4.675***	14.844
	(3.284)	(−1.669)	(4.605)	
$OECD_{dummy}$	0.887***	−0.887	1.516*	9.331
	(2.850)	(−1.200)	(1.762)	
C72	1.220***	−12.100**	−2.294	9.758
	(3.085)	(−2.262)	(−0.390)	
CFNAI	1.331***	0.313	−0.293	7.277
	(3.661)	(0.893)	(−1.122)	
USRPs	1.004***	−0.035**	0.050***	15.359
	(3.199)	(−1.673)	(4.475)	

This table reports the in-sample predictive performance of the ESG market indicator over the economic cycle. The regression is given as follows,

$$R_{t+1} = \alpha + \beta_{ESG} ESG_t + \beta_{EC} EC_t + \beta_{ESG\times EC} ESG_t \cdot EC_t + \epsilon_{t+1}$$

where R_{t+1} is the future value of the excess return of the S&P 500 index and ESG_t is the ESG market indicator. EC is a measure of the economic cycle. The measures of the economic cycle are NBER and OECD recession dummies, which equal 1 during a recession and 0 otherwise. We also use the consumption cycle (C72) in Atanasov et al. (2020) to represent the continuous economic cycle. CFNAI is the Chicago Fed National Activity Index from the Federal Reserve Bank of Chicago. USRPs are U.S. recession probabilities which are derived from a dynamic-factor Markov-switching model in Chauvet and Piger (2008) using four monthly indicators: non-farm payroll, industrial production index, real personal income without transfers, and real manufacturing/trade sales.
We use data from the Federal Reserve Bank of St. Louis—the data range from July 2007 to December 2021. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

Table 9
ESG, market volatility, firm fundamentals, and economic activity.

	β_1	t-statistics	β_2	t-statistics	$adj-R^2$
	Panel A: Market Volatility				
Market Volatility	−0.102 **	−2.335	−0.356 ***	−4.589	15.192
	Panel B: Firm Fundamentals				
ROA	-0.317×10^{-3} *	−1.724	0.080**	2.543	1.335
EBIT	−0.018	−0.904	0.039	0.680	−0.629
GPM	-0.489×10^{-3}**	−2.087	−0.017	−0.729	−0.223
OPMAD	-0.719×10^{-3}*	−1.798	0.012	0.643	0.252
GPTA	-0.577×10^{-3}**	−1.977	0.086**	2.340	2.220
CFM	-0.435×10^{-3}	−1.173	0.007	0.187	−0.809
DY	−0.044**	−2.155	0.067	0.646	5.426
BM	−0.005***	−3.972	−0.081	−1.348	4.091
PS	0.039***	3.157	−0.094	−0.834	8.057
PEOP	0.395**	2.413	0.059	1.549	1.839
CAPEI	13.079*	1.818	−0.035	−1.395	6.083
FCFOCF	0.001	0.240	−0.024	−1.303	−1.110
CFTD	0.374×10^{-3}	0.709	−0.027	−0.526	−0.230
TDE	−0.104	−0.206	−0.002	−0.335	−1.148
LDBE	−0.014	−1.160	−0.084	−1.006	0.430
SIC	−0.003*	−1.693	0.031	0.978	−0.639
AT	-0.693×10^{-3}	−0.960	0.057	1.068	0.055
IT	−0.324	−0.294	0.000	−0.284	−1.163
AAA	-0.139×10^{-3}	−0.624	0.097***	3.263	0.057
	Panel C: Economic Activity				
INDPRO	−0.209	−1.332	0.231**	2.039	6.067
PMI	−0.086	−0.256	−0.175**	−2.038	1.980
USNFPC	-397.000×10^{3}	−1.233	−0.428**	−2.024	19.012
RSI	0.109	0.450	0.093	0.688	−0.262
IS	0.603×10^{-3}	0.321	0.028	0.218	−1.070

This table reports the results of the following regression.

$$\Delta Dependent_{t+1} = \beta_0 + \beta_1 ESG_t + \beta_2 \Delta Dependent_t + \epsilon_{t+1}$$

where *Dependent* serves as a measure of market volatility in Panel A, firm fundamentals in Panel B, and economic activity in Panel C, respectively. *ESG* is the ESG market indicator, and β_1 is the coefficient that we are interested in. It represents the predictive power of the ESG market indicator on the dependent variables. Because most of the dependent variables are persistent, we control for them in the regression, and β_2 denotes the coefficient for each. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

where *FF* represents a measure of firm fundamentals. We incorporate a series of widely recognized firm fundamental variables from the literature (De Roon and Szymanowska 2012, Dickinson 2011, D'Mello and Miranda 2010, Fairfield and Yohn 2001, Faulkender et al. 2012, Feng et al. 2015, Han et al. 2022, Lee et al. 2009, Lev and Nissim 2006, Rozenbaum 2019 and among others). The variables are listed below, categorized for clarity:

- Profitability Ratios:
  - Return on Assets (ROA): Measures the efficiency in generating profit relative to total assets.
  - Earnings Before Interest and Taxes (EBIT): Indicates the company's core profitability before financing costs.
  - Gross Profit Margin (GPM): Shows the percentage of revenue exceeding the cost of goods sold.
  - Operating Profit Margin After Depreciation (OPMAD): Reflects profitability after accounting for depreciation.
  - Gross Profit to Total Assets Ratio (GPTA): Assesses profitability relative to total assets.
  - Cash Flow Margin (CFM): Measures cash flow efficiency relative to sales.
  - Dividend Yield (DY): Reflects the annual dividends paid relative to the stock price.
- Valuation Ratios:
  - Book-Market Ratio (BM): Compares book value to market value, indicating company valuation.
  - Price to Sales Ratio (PS): Compares market price to sales revenue, assessing company valuation.
  - Price to Operating Earnings Ratio (PEOP): Indicates the price investors pay per unit of operating earnings.
  - Shiller's Cyclically Adjusted P/E Ratio (CAPEI): Adjusted P/E ratio accounting for cyclical fluctuations.
- Financial leverage:
  - Free Cash Flow to Operating Cash Flow (FCFOCF): Examines cash generation versus operational cash flow.
  - Cash Flow to Total Debt (CFTD): Indicates the company's ability to cover debt with cash flow.
  - Total Debt to Equity (TDE): Measures the level of debt financing relative to equity.
  - Long-Term Debt to Book Equity (LDBE): Measures long-term debt as a proportion of book equity.
- Efficiency Ratios:
  - Sales to Invested Capital (SIC): Shows revenue generated per unit of invested capital.
  - Asset Turnover (AT): Indicates how efficiently assets are used to generate sales.
  - Inventory Turnover (IT): Assesses how frequently inventory is sold and replaced.
  - Accruals to Average Assets (AAA): Measures the level of accruals relative to average assets.

Data for these variables are sourced from the Financial Ratios Suite tool, provided by Wharton Research Data Services (WRDS), and cross-sectional averages are calculated based on S&P 500 companies.

The predictive results are presented in Panel B of Table 9. The findings reveal that a key characteristic of our ESG signal is its strong and consistent link to future changes in both firm profitability and valuation. The coefficients for ESG_t in regressions predicting future profitability measures are negative and mostly significant at the 5% level, indicating that a higher ESG market indicator is associated with a decline in profitability measures in the following month. This finding aligns with research demonstrating the nuanced relationship between ESG incidents and financial performance, where certain ESG incidents may lead to reduced short-term profitability as firms incur higher costs for dealing with ESG incidents (Albuquerque et al., 2019; Khan et al., 2016).

For valuation measures, the coefficients for ESG_t are all significant at the 5% level. A higher ESG indicator level predicts a future decline in valuation based on dividend yield (DY) and book-to-market (BTM), alongside a predicted increase in price-to-sales (PS) and price-to-earnings (PEOP), suggesting a complex repricing dynamic following the news captured by our signal. In contrast, the ESG coefficients in regressions predicting changes in all financial leverage and efficiency measures are not statistically significant.

Overall, these results characterize our ESG predictive signal as being most informative about future shifts in aggregate corporate profitability and market valuation. This provides a direct economic underpinning for the indicator's ability to predict stock returns, demonstrating that the signal is not disconnected from the real economy but instead a leading indicator of its fundamental health.

5.5. Distinguishing the signal from macroeconomic activity

Finally, we characterize our ESG predictive signal by testing whether it is simply a proxy for future macroeconomic activity. It is plausible that a market-wide increase in reported ESG incidents could reflect or precede a broader downturn in the business cycle. If our signal primarily captures this kind of information, its ability to predict stock returns might simply reflect the well-known link between the business cycle and the equity premium. We therefore test the hypothesis that our ESG indicator is a leading indicator for future changes in broad macroeconomic activity.

Continuing with the robust 'level-on-change' framework used throughout this section, we test whether our ESG indicator (ESG_t) can predict future one-period changes (ΔREC_{t+1}) in several standard macroeconomic variables. The model specification is

$$\Delta REC_{t+1} = \beta_0 + \beta_1 ESG_t + \beta_2 \Delta REC_t + \epsilon_{t+1} \tag{14}$$

where *REC* represents a measure of economic activities. Followed to Li et al. (2023), we consider various economic

indicators: the industry productivity (IP), Chicago Fed National Activity Index (CFNAI), U.S. recession probability (USRPs), Chicago Purchasing Managers' Index (PMI), U.S. non-farm payroll changes (USNFPC), Retail Sales Index (RSI), and inventories/sales (IS). The data for PMI, USNFPC, and RSI are from Datastream, and IS data is sourced from the Federal Reserve Bank of St. Louis.

Panel C of Table 9 presents the results. The findings show a consistent and clear pattern: our ESG predictive signal does not appear to be a significant leading indicator for future changes in the macroeconomic variables we test. Across all specifications, the coefficient on our ESG indicator (β_1) is statistically insignificant at conventional levels.

This null result is highly informative. It strongly suggests that the predictive power of our ESG signal for stock returns does not operate through a channel in which it first predicts broad business-cycle movements. This finding stands in sharp contrast to our results for market volatility and firm fundamentals. Taken together, our analysis indicates that the economic information captured by our PLS-based signal is more closely tied to financial market risk perceptions and firm-specific performance expectations than it is to aggregate economic activity. This helps to isolate the specific nature of the predictability we have uncovered.

6. Robustness checks

6.1. Transaction costs

In our initial analysis, transaction costs are not considered. However, it is well recognized that transaction costs are an intrinsic part of equity markets and can affect portfolio performance. To evaluate the implications for our primary findings, this section incorporates transaction costs at rates of 10, 30, and 50 basis points.

Table 10 reports the economic values associated with the ESG market indicator under various levels of transaction costs as investors allocate their capital between the equity market portfolio and a risk-free asset. Assuming a risk aversion coefficient of 3 for an investor and an optimal weight on the market portfolio ranging from -1 to 2, the annualized CER gain decreases from 13.870% in the absence of transaction costs to 12.998%, 11.245%, and 9.481% for transaction costs of 10, 30, and 50 basis points, respectively. Meanwhile, investment strategies based on the ESG market indicator result in a decrease in the annual Sharpe ratio, declining from 1.306 to 1.268, 1.192, and 1.115. When investors exhibit higher risk aversion, rising transaction costs are associated with lower CER gains and lower Sharpe ratios. Nevertheless, our ESG-centered investment strategies consistently offer substantial economic benefits to mean–variance investors, even when accounting for transaction costs.

6.2. Alternative risk aversion

In our prior analysis, we assumed that the mean–variance investor has a relative risk aversion coefficient (γ) of 3. The existing literature indicates that the economic value of investment strategies might fluctuate across different levels of this coefficient. To assess the robustness of our findings to variations in this parameter, we extend our investigation to include two additional plausible risk aversion coefficients: $\gamma = 5$ and 7.

Table 10 presents the results for economic values under these alternative risk-aversion settings. Assuming an optimal weight allocation to the market portfolio ranging from -1 to 2 and accounting for transaction costs, we observe that a decrease in risk aversion during our sample period increases the CER gains associated with our ESG strategy. For example, with a transaction cost of 50 basis points, the CER gain increases from 4.269% when $\gamma = 7$, to 8.985% when $\gamma = 5$, and further to 9.481% when $\gamma = 3$. In addition, the Sharpe ratios remain relatively stable for γ values of 5 and 7 under the same transaction cost scenarios, exceeding the ratio observed when the risk aversion coefficient is set at 3. These variations in economic values result from shifts in risk aversion, leading investors to potentially reallocate assets towards relatively safer options, in this context, risk-free Treasury assets, optimizing their utility. Our findings confirm that the predictive ability of the ESG market indicator remains robust even as investor risk aversion increases, thereby

Table 10
Alternative transaction costs, risk aversion and weight constraints.

γ	Transaction cost	CER Gain	Sharpe Ratio
$\omega^* = [-1, 2]$			
3	0	13.870	1.306
5	0	14.293	1.396
7	0	10.066	1.380
3	0.1%	12.998	1.268
5	0.1%	13.239	1.348
7	0.1%	8.913	1.323
3	0.3%	11.245	1.192
5	0.3%	11.119	1.250
7	0.3%	6.598	1.210
3	0.5%	9.481	1.115
5	0.5%	8.985	1.152
7	0.5%	4.269	1.096
$\omega^* = [0, 1]$			
3	0	8.988	1.320
5	0	12.748	1.443
7	0	11.786	1.483
3	0.1%	8.510	1.293
5	0.1%	12.163	1.407
7	0.1%	11.089	1.439
3	0.3%	7.552	1.239
5	0.3%	10.990	1.336
7	0.3%	9.690	1.351
3	0.5%	6.589	1.185
5	0.5%	9.810	1.264
7	0.5%	8.284	1.262

This table reports the out-of-sample economic performance under various settings of investor risk aversions and transaction costs. γ is the investor's constant risk aversion. ω^* is the optimal weight on the market portfolio. "CER Gain" is the difference of the certainty equivalent return (CER) between predictive models and the benchmark model. "Sharpe Ratio" is calculated as the annualized average excess return of a portfolio divided by its annualized standard deviation. See Section 4.2 for further details. The initial training window for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021.

affirming the resilience of our asset allocation strategies across diverse levels of risk aversion.

6.3. Alternative weight constraints

In our earlier analysis, we postulated that the optimal weight for the market portfolio, denoted by ω^*, lies between -1 and 2. This implies that investors can engage in short sales up to 100% or use financial leverage through margin borrowing up to 100%. In this part, we restrict the weight to the range 0 to 1. This adjustment aims to eliminate the possibility of short selling and does not allow for financial leverage.

The results are presented in Table 10. The upper panel displays the results related to an aggressive strategy ($\omega^* = [-1, 2]$), whereas the lower panel outlines those tied to a more conservative approach ($\omega^* = [0, 1]$). The CER gains for both strategies are predominantly greater than those of the benchmark model, and their Sharpe ratios are all above 1. These results suggest that the economic benefits derived from the ESG market indicator remain significant and robust across different trading styles. In most cases, the aggressive strategy yields higher CER gains than the conservative approach, a plausible outcome given the permission for short selling and greater financial leverage. Conversely, the Sharpe ratios associated with the conservative strategy tend to exceed those of the aggressive strategy, largely due to the conservative approach's substantially lower return volatility.

6.4. The international transmission of the U.S. ESG signal

While our primary analysis focuses on the U.S. market due to data availability, a crucial question is whether the risks captured by our U.S.-based ESG indicator have international relevance. A large body of literature, pioneered by Rapach et al. (2013), demonstrates that U.S. financial variables often have significant predictive power for international equity markets, suggesting a strong transmission channel for U.S.-based information and risk.

Given that we have established our U.S. ESG indicator as a significant predictor of U.S. market returns, we now test whether it can also directly forecast future returns in other developed and emerging markets. If so, this would suggest that the indicator captures not just a local phenomenon, but a globally priced systematic risk that is first reflected in the U.S.

We conduct an out-of-sample forecasting exercise for 38 international stock markets, using our single U.S.-based ESG predictive indicator as the sole predictor for each country's future market return. The results are presented in Table 11. Our ESG signal exhibits statistically significant out-of-sample predictive power in approximately a quarter of global markets. The strongest results are observed in countries with close financial and economic ties to the U.S., particularly developed economies such as Canada ($R^2_{OS} = 10.33\%$), Australia ($R^2_{OS} = 7.92\%$), and Finland ($R^2_{OS} = 7.95\%$). We also find significant predictability in several smaller European markets, such as Estonia, Latvia, and Slovenia, as well as in countries from other regions, including Israel and Mexico.

Conversely, the indicator does not show predictive ability for other countries, including major Asian markets such as Japan, China, and India, as well as several large Eurozone countries. This heterogeneity suggests that the international transmission of U.S.-based ESG risk is conditional rather than universal. It likely depends on factors such as financial market integration, investor base composition, ESG regulatory alignment, and similarities in industrial structure.

Overall, this analysis provides initial evidence that our U.S.-based ESG indicator captures more than just a domestic signal. Its significant predictive power for a subset of international markets, particularly those with close economic ties to the U.S., suggests the existence of an international transmission channel for these specific ESG risks. Fully understanding the factors that determine when and where this transmission occurs is a topic that merits further research.

7. Conclusion

In the academic literature, there has been limited exploration of how market-based ESG information affects future aggregate equity markets—our study endeavors to bridge this research gap. First, we collect firm-level ESG incident data from RepRisk covering January 2007 to December 2021, and employ the PLS method to develop an ESG market indicator incorporating 28 specific incident issues. To the best of our knowledge, this indicator exhibits the strongest predictability for S&P 500 excess market returns in both in-sample and out-of-sample periods among the existing literature.

Our findings reveal that the ESG market indicator has both statistically and economically significant predictive power on excess market returns, as confirmed by both in-sample and out-of-sample analyses. Notably, the forecasting performance of the ESG market indicator remains robust and significant even after controlling for commonly considered macroeconomic variables, the consumption cycle variable, and uncertainty variables. We validate our results through a series of robustness checks that consider alternative transaction costs, varying degrees of risk aversion, stricter weight constraints on the market portfolio, and the impact of economic cycles. This study proposes that a risk-averse investor does not need to establish an ESG portfolio by selecting stocks from thousands of individual stocks. Instead, she can easily realize substantial benefits by distributing assets between a risk-free asset and the market portfolio. Predictions from our incident-based ESG market indicator guide the strategy.

We also explore the underlying economic mechanisms contributing to the return predictability of the ESG market indicator. We find that cash flow and discount rate news are key drivers of ESG return predictability. Moreover, the predictive ability of our indicator also partially arises from its capacity to forecast market volatility and firm fundamentals; however, it is unlikely to arise from its ability to predict economic activity.

Following Welch and Goyal (2008) and Goyal et al. (2024), continuing to evaluate established predictors using newly available data provides the most rigorous test

Table 11
International predictability of US ESG signals.

Country	R^2_{OS}	MSFE-adjusted	CER Gain	Sharpe Ratio
Australia	7.922**	2.034	20.782	1.228
Austria	1.297	1.041	−1.359	0.225
Belgium	−10.084	−1.120	−20.247	−0.416
Brazil	−0.748	0.967	−6.116	0.103
Canada	10.330***	2.604	19.556	1.153
China	−9.772	0.403	−2.447	0.066
Colombia	−0.333	0.769	6.801	0.391
Czech Republic	0.451	0.878	−1.582	0.316
Denmark	1.727	1.256	−0.924	1.047
Estonia	7.965**	2.122	26.623	1.184
Finland	7.946**	2.275	10.615	0.873
France	0.024	1.064	3.964	0.641
Germany	−1.037	0.915	−0.134	0.409
Greece	−14.232	−0.482	3.512	0.031
Iceland	4.039**	2.349	16.454	0.340
India	−16.355	0.051	−0.658	0.309
Ireland	2.124	1.158	2.722	0.383
Israel	6.831***	2.436	19.655	1.214
Italy	−3.829	0.016	−7.079	−0.077
Japan	−4.990	0.114	−4.824	0.141
Latvia	2.822***	2.620	11.568	1.418
Lithuania	0.017*	1.585	1.173	0.872
Luxembourg	−15.006	−0.688	−11.655	−0.549
Mexico	3.636**	2.343	11.776	0.756
Netherlands	−0.448	0.734	1.414	0.614
New Zealand	3.260*	1.649	2.464	0.759
Norway	4.392**	1.923	10.320	0.831
Poland	−4.065	0.420	−16.496	−0.319
Portugal	−10.202	−0.891	−6.039	−0.501
Slovakia	−10.865	0.033	−5.494	−0.046
Slovenia	1.808**	1.881	20.483	0.603
South Africa	−8.702	−0.090	−7.955	−0.165
South Korea	−0.062	1.132	6.951	0.545
Spain	−11.727	−1.528	−17.680	−0.657
Sweden	0.109	1.301	6.426	1.077
Switzerland	−0.636	0.587	0.508	0.802
Turkey	−5.846	0.066	−2.684	0.140
United Kingdom	−1.249	0.606	5.509	0.353

This table reports the out-of-sample forecasting performance of the US ESG-based predictor on 38 international markets. The initial training window for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

of their robustness. As our dataset ends in 2021, extending the analysis with post-2021 data would therefore be a valuable avenue for future research.

CRediT authorship contribution statement

Zhiyong Li: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Weiping Qin:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis.

Data and code availability

The data used in this study are obtained from WRDS and Datastream and are subject to third-party licensing restrictions; they are therefore not publicly available. Researchers with appropriate access can obtain the data directly from the providers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. RepRisk incident classification and definition of traditional predictors

Fig. A.1 plots the time series for each of these four component indicators. A visual inspection reveals both commonalities and notable differences in their dynamics. While all four components exhibit a general increase in volatility during the latter half of the sample, their underlying behavior is clearly differentiated. The Social indicator exhibits low and stable volatility before 2013, followed by a sustained increase in the magnitude of its fluctuations thereafter. In contrast, the Governance indicator is characterized by relative stability, punctuated by a few large, transient shocks, most notably a major positive spike in mid-2017 and a sharp negative spike in early 2019. The Environmental and Cross-cutting indicators both display more persistent, cyclical fluctuations throughout the sample, though they are also marked by sharp, isolated spikes. This heterogeneity provides strong support for our modeling strategy, as it indicates that each

Table A.1
RepRisk Incident Classification and Frequency.

Categories	Issues	Frequency	Percent
Environmental	Animal mistreatment	863	0.54%
	Climate change, GHG emissions, and global pollution	6632	4.16%
	Impacts on landscapes, ecosystems, and biodiversity	11,894	7.46%
	Local pollution	9164	5.74%
	Overuse and wasting of resources	1044	0.65%
	Waste issues	3198	2.00%
	Total	32,795	20.56%
Social	Child labor	1723	1.08%
	Discrimination in employment	2479	1.55%
	Forced labor	2370	1.49%
	Freedom of association and collective bargaining	1861	1.17%
	Human rights abuses, corporate complicity	10,762	6.75%
	Impacts on communities	13,344	8.36%
	Local participation issues	2017	1.26%
	Occupational health and safety issues	4998	3.13%
	Poor employment conditions	6874	4.31%
	Social discrimination	1263	0.79%
	Total	47,691	29.90%
Governance	Anti competitive practices	4878	3.06%
	Corruption, bribery, extortion, money laundering	5,307	3.33%
	Executive compensation issues	1383	0.87%
	Fraud	7942	4.98%
	Misleading communication	4447	2.79%
	Tax evasion	1242	0.78%
	Tax optimization	2512	1.57%
	Total	27,711	17.37%
Cross-cutting	Controversial products and services	7705	4.83%
	Products (health environmental issues)	8253	5.17%
	Supply chain issues	9670	6.06%
	Violation of international standards	1650	1.03%
	Violation of national legislation	24,004	15.05%
	Total	51,282	32.15%
	Other ESG issues	50	0.03%
Total		159,529	100%

Table A.2
Traditional predictors for stock returns.

Predictor	Abbreviation	Brief Description
Dividend-payout ratio	LNDE	The logarithm of a 12-month moving sum of dividends on the S&P 500 index, subtracted by the logarithm of a 12-month moving sum of earnings
Long-term yield	LTY	The long-term government bond yield
Dividend yield	LNDY	The logarithm of a 12-month moving sum of dividends on the S&P 500 index, subtracted by the logarithm of lagged stock prices
Dividend price ratio	LNDP	The logarithm of a 12-month moving sum of dividends on the S&P 500 index, subtracted by the logarithm of stock prices.
Treasury bill rate	TBL	The interest rate on a 3-month Treasury bill (secondary market)
Earnings price ratio	LNEP	The logarithm of a 12-month moving sum of earnings on the S&P 500 index, subtracted by the logarithm of stock prices
Book-to-market ratio	BM	The ratio of book value divided by market value for the Dow Jones Industrial Average
Inflation	INFL	The consumer price inflation (CPI) for all urban consumers
Default yield spread	DFY	The difference between Moody's BAA- and AAA-rated corporate bond yields
Net equity expansion	NTIS	The ratio of a 12-month moving sum of net equity issues by NYSE-listed stocks to the total end-of-year market capitalization of NYSE stocks
Term spread	TMS	The long-term yield minus the Treasury bill rate
Long-term return	LTR	The returns on long-term government bonds
Default return spread	DFR	The long-term corporate bond return minus the long-term government bond return
Stock return variance	SVAR	The sum of squared daily returns on the S&P 500 index
Total real uncertainty	TRU	The measurement of total real uncertainty developed by Jurado et al. (2015)
Total financial uncertainty	TFU	The measurement of total financial uncertainty developed by Jurado et al. (2015)
Total macroeconomic uncertainty	TMU	The measurement of total macroeconomic uncertainty developed by Jurado et al. (2015)
Economic policy Uncertainty	EPU	The measurement of economic policy uncertainty introduced by Baker et al. (2016)
VIX index	VIX	The measurement of the market's expectation of stock market volatility over the next 30-day period conveyed by S&P 500 Index options, created by the Chicago Board Options Exchange (CBOE).
Consumption cycle	C72	The consumption cycle introduced by Atanasov et al. (2020)

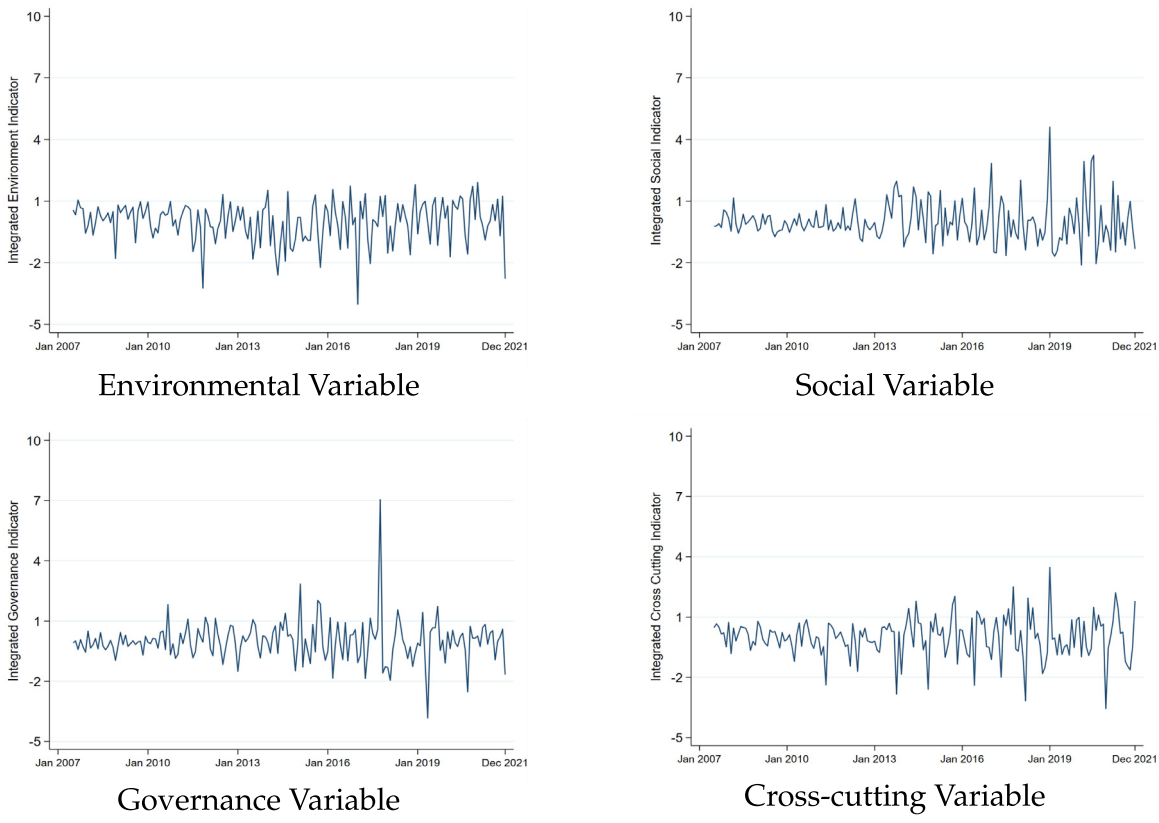


Fig. A.1. ESG incident components.
This figure plots the time series of the integrated ESG components over the sample period from July 2007 to December 2021. The four components are environmental, social, governance, and cross-cutting variables. All variables are standardized and detrended.

dimension captures a distinct informational channel of ESG-related risk. It further highlights the value of the PLS method, which can optimally weight these complementary signals rather than treating them as homogeneous inputs.

Appendix B. In-sample and out-of-sample performance of ESG individual indicators

See Tables B.3 and B.4.

Appendix C. Partial Least Square (PLS) method and ESG market indicator construction

Kelly and Pruitt (2015) introduce the three-pass regression filter (3PRF) method and demonstrate that the PLS method is a special case of 3PRF. In this study, we adopt the PLS method using the 3PRF framework because the 3PRF approach to indicator construction is more economically intuitive. Specifically, our construction satisfies the following three conditions: "(i) the predictors are demeaned and variance-standardized in a preliminary step, (ii) the first two regression passes are run without constant terms, and (iii) proxies are automatically selected". Consequently, our method is effectively the PLS method. We consider an unobserved ESG market indicator that is assumed to forecast future stock returns. This can be formally expressed as:

$$R_{t+1} = \alpha_1 + \beta ESG_t + \epsilon_{t+1} \tag{C.1}$$

where, R_{t+1} is the excess market returns at month $t+1$, ESG_t is the unobserved ESG market indicator at month t, α_1 is a constant, and ϵ_{t+1} is an unpredictable shock at month $t+1$, independent of the ESG indicator.

Suppose there are N ESG individual indicators, labeled as $esg_1, esg_2, \ldots, esg_n$. Each of these indicators partially maps the true ESG market indicator, albeit with certain errors. Therefore, the relationship between the ESG market indicator ESG and each individual indicator esg can be described by the following regression:

$$esg_{i,t} = \alpha_2 + \theta_i ESG_t + \eta_t e_t + \nu_t \tag{C.2}$$

where, θ_i denotes the loading of the ESG market indicator on ith individual indicator. The term e_t is the common error applicable to all individual indicators that are not relevant to market returns, and ν_t is an idiosyncratic error.

According to the Eqs. (C.1) and (C.2), it is likely that individual indicators correlate with future stock returns. To estimate the sensitivities of the individual indicators to returns, the following regression is used:

$$esg_{i,t} = \alpha_3 + \lambda_i R_{t+1} + \mu_t \tag{C.3}$$

Table B.3
In-sample predictive regression of ESG individual indicators.

	β	t	$Adj-R^2(\%)$
Animal mistreatment (IE1)	0.032	0.429	−0.459
Climate change, GHG emission, and global pollution (IE2)	−0.002	−0.152	−0.574
Impacts on landscapes, ecosystems, and biodiversity (IE3)	−0.028**	−2.493	1.752
Local pollution (IE4)	−0.025*	−1.703	0.758
Overuse and wasting (IE5)	−0.079	−1.441	0.592
Waste issues (IE6)	−0.028	−0.974	−0.115
Child labor (IS7)	−0.025	−0.698	−0.373
Discrimination in employment (IS8)	0.008*	1.737	−0.355
Forced labor (IS9)	0.034	0.904	0.011
Freedom of association and collective bargaining (IS10)	0.030	0.814	−0.315
Human rights abuses, corporate complicity (IS11)	−0.006	−0.701	−0.362
Impacts on communities (IS12)	−0.018*	−1.875	0.401
Local participation issues (IS13)	0.000	−0.004	−0.585
Occupational health (IS14)	0.020	0.793	−0.17
Poor employment conditions (IS15)	0.006	1.071	−0.455
Social discrimination (IS16)	0.111**	2.336	1.915
Anti competitive practices (IG17)	−0.020	−0.891	−0.291
Corruption, bribery, extortion, money laundering (IG18)	−0.042**	−2.172	2.014
Executive compensation issues (IG19)	0.018	0.779	−0.291
Fraud (IG20)	−0.009	−0.542	−0.483
Misleading communication (IG21)	0.005	0.818	−0.497
Tax evasion (IG22)	0.005	0.815	−0.536
Tax optimization (IG23)	0.001	0.231	−0.579
Controversial products and services (IC24)	−0.022*	−1.920	0.940
Products (health environmental issues) (IC25)	−0.010	−0.627	−0.423
Supply chain issues (IC26)	0.001	0.027	−0.584
Violation of international standards (IC27)	−0.036*	−1.751	0.353
Violation of national legislation (IC28)	−0.002	−0.365	−0.551
Other ESG issues (IO29)	0.172	0.607	−0.363

This table reports the in-sample forecasting performance of issue-based ESG individual indicators using single-predictor predictive regression, where the future value of the excess return of the S&P 500 index is the dependent variable and one of the ESG individual indicators is the independent variable. The data range from July 2007 to December 2021. Newey–West t-values are reported in the table. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

Now, if we let $\hat{\lambda}_i$ serve as the independent variable and $esg_t = [esg_{1,t}\ esg_{2,t}\ \ldots\ esg_{n,t}]$ as the dependent variable, we can run the following cross-sectional regression for each period:

$$esg_t = \hat{ESG}_t \lambda_i + \tau_t \tag{C.4}$$

The coefficient $\hat{ESG}_t$ represents the estimate of the ESG market indicator at month t.

Using individual ESG indicators, we can construct an ESG market indicator. On the other side, the ESG market indicator can be considered as a linear combination of ESG individual indicators according to the following equation:

$$ESG_t = \sum \pi_i esg_{i,t} \tag{C.5}$$

Here, π_i represents the contribution of each ESG individual indicator to the ESG market indicator. The covariance between the ESG individual indicator and future excess returns determines π_i. Table C.5 presents the weight (πs) of issue-based ESG individual indicators to the ESG market indicator over the full sample. We can see that around half of the π_is are positive because these individual indicators are positively related to future equity returns. The positive π_is also indicate that increases in these individual indicators lead to higher values of the ESG market indicator, and vice versa.

Appendix D. Decompose return innovations into discount rate news and cash flows news

Many valuation models in the literature suggest that asset prices should equal the sum of discounted future cash flows and thus be determined by both future expected cash flows and discount rates (e.g., Cochrane 2011). This concept can be formally presented as follows:

$$R_{t+1} - E_t(R_{t+1}) = \underbrace{(E_{t+1} - E_t)\sum_{k=0}^{\infty} \rho^k \Delta d_{t+1+k}}_{CFN_{t+1}} - \underbrace{(E_{t+1} - E_t)\sum_{k=1}^{\infty} \rho^k R_{t+1+k}}_{DRN_{t+1}} \tag{D.6}$$

where E is the expectation operator, Δd is the growth rate of dividends, and ρ is the discount rate, which can be computed using the subsequent equation,

$$\rho = \frac{1}{1 + e^{\overline{DP}}} \tag{D.7}$$

where $\overline{DP}$ is the average of the logged dividend price ratios. The first term on the right-hand side is defined as the cash flow news (CFN_{t+1}) and the second term is the discount rate news (DRN_{t+1}).

Table B.4
Out-of-sample predictive regression of ESG individual indicators.

	R^2_{os}	*MSFE − adjusted*
Animal mistreatment (IE1)	−0.608	0.026
Climate change, GHG emission, and global pollution (IE2)	−0.271	−0.600
Impacts on landscapes, ecosystems, and biodiversity (IE3)	2.624**	1.769
Local pollution (IE4)	1.021	1.000
Overuse and wasting (IE5)	2.125*	1.547
Waste issues (IE6)	0.300	0.407
Child labor (IS7)	−0.255	−0.590
Discrimination in employment (IS8)	−1.680	1.150
Forced labor (IS9)	1.994	1.242
Freedom of association and collective bargaining (IS10)	2.000**	1.848
Human rights abuses, corporate complicity (IS11)	−4.781	−0.802
Impacts on communities (IS12)	−0.0350	0.060
Local participation issues (IS13)	−0.543	−0.266
Occupational health (IS14)	3.948**	1.783
Poor employment conditions (IS15)	1.290*	1.551
Social discrimination (IS16)	7.851**	1.824
Anti competitive practices (IG17)	−0.548	−0.709
Corruption, bribery, extortion, money laundering (IG18)	2.258	1.119
Executive compensation issues (IG19)	0.834**	1.855
Fraud (IG20)	−2.675	−1.264
Misleading communication (IG21)	1.093*	1.432
Tax evasion (IG22)	−1.435	−0.891
Tax optimization (IG23)	0.097	0.433
Controversial products and services (IC24)	2.232**	1.978
Products (health environmental issues) (IC25)	−0.922	−0.116
Supply chain issues (IC26)	0.324	0.468
Violation of international standards (IC27)	1.261**	1.863
Violation of national legislation (IC28)	0.523	0.910
Other ESG issues (IO29)	0.142	0.877

This table reports the out-of-sample forecasting performance of issue-based ESG individual indicators. The indicators are obtained by Eq. (1). R^2_{OS}, introduced by Campbell and Thompson (2008), is the percentage reduction in mean squared forecast error of the predictive regression in comparison to the benchmark model, calculated according to Eq. (4). *MSFE − adjusted*, introduced by Clark and West (2007), is the statistics for testing hypothesis $H_0 : R^2_{OS} \leq 0$ against $H_A : R^2_{OS} > 0$. The initial training period for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

Table C.5
Issue-based ESG individual indicator weights on the ESG market indicator.

Individual Indicator	π	Individual Indicator	π
Animal mistreatment (IE1)	0.050	Social discrimination (IS16)	0.125
Climate change, GHG emission, and global pollution (IE2)	−0.119	Anti competitive practices (IG17)	0.023
Impacts on landscapes, ecosystems, and biodiversity (IE3)	−0.399	Corruption, bribery, extortion, money laundering (IG18)	−0.296
Local pollution (IE4)	−0.201	Executive compensation issues (IG19)	0.096
Overuse and wasting of resources (IE5)	−0.081	Fraud (IG20)	0.098
Waste issues (IE6)	−0.121	Misleading communication (IG21)	0.139
Child labor (IS7)	−0.011	Tax evasion (IG22)	0.102
Discrimination in employment (IS8)	0.208	Tax optimization (IG23)	0.127
Forced labor (IS9)	0.213	Controversial products and services (IC24)	−0.324
Freedom of association and collective bargaining (IS10)	0.051	Products (health environmental issues) (IC25)	−0.051
Human rights abuses, corporate complicity (IS11)	−0.097	Supply chain issues (IC26)	0.233
Impacts on communities (IS12)	−0.211	Violation of international standards (IC27)	−0.079
Local participation issues (IS13)	−0.039	Violation of national legislation (IC28)	0.428
Occupational health and safety issues (IS14)	0.244	Other ESG issues (IO29)	−0.001
Poor employment conditions (IS15)	0.191		

This table shows the contribution (π) of each ESG individual indicator to the ESG market indicator. The detailed procedure is described in Appendix C in the appendix. The whole period spans from July 2007 to December 2021.

Therefore, we can rewrite Eq. (D.6) as:

$$R_{t+1} = E_t(R_{t+1}) + CFN_{t+1} - DRN_{t+1} \tag{D.8}$$

From Eq. (D.8), it becomes evident that the unexpected market returns can be decomposed into three distinct components: the expected future return, the cash flow news, and the discount rate news. If the ESG market indicator successfully predicts future market returns, it should be able to forecast at least one of these components.

To perform the return decomposition, we employ a first-order Vector Autoregression (VAR) framework, developed by Campbell (1991) and Campbell and Vuolteenaho (2004), as stated below:

$$Y_{t+1} = BY_t + \epsilon_{t+1} \tag{D.9}$$

where B represents the coefficients matrix and Y is a vector of variables, including stock returns and macroeconomic predictors.

Based on the established VAR framework, the cash flow news and discount rate news can be estimated and expressed as follows:

$$\begin{aligned} CFN_{t+1} &= a'\epsilon_{t+1} + a'\rho B(I-\rho B)^{-1}\epsilon_{t+1} \\ DRN_{t+1} &= a'\rho B(I-\rho B)^{-1}\epsilon_{t+1} \end{aligned} \tag{D.10}$$

where, $a' = [1 \;\; 0 \ldots 0]_{1\times n}$, I is an identity matrix, n is the number of variables.

Chen and Zhao (2009) have demonstrated that this decomposition is highly sensitive to the choice of predictors. To address this concern, we adopt strategies from the approaches of Rapach et al. (2016) and Chu et al. (2022) to broaden our array of variables for the VAR model. Initially, we consider the variables identified by Campbell and Vuolteenaho (2004), including bond yield spread; Shiller's price-to-earnings ratio for the S&P 500 index; and the small-stock value spread (defined as the difference between book-to-market ratios of small value and small growth stocks). Subsequently, as suggested by Chen and Zhao (2009), we employ principal component analysis (PCA) to obtain the first five components from a range of macroeconomic variables. These include term spread, Shiller's price-to-earnings ratio, small-stock value spread, credit spread, long-term corporate bond return, book-to-market ratio, 3-month T-bill rate, inflation, corporate issuing activity, and stock variance. Existing literature underlines that the dividend-price ratio plays a vital role in estimating both cash flow news and discount rate news (e.g. Rapach et al. 2016). Therefore, one of our VAR model specifications integrates log excess market returns, the log dividend-price ratio, and/or one of the remaining 13 macroeconomic variables identified by Welch and Goyal (2008). Finally, we apply PCA to extract the first principal component from these 13 macroeconomic variables. We then incorporate this component into our VAR framework, alongside the log market return and log dividend-price ratio.

Appendix E. Multivariate out-of-sample predictive analysis

The predictive ability of the ESG market indicator has been comprehensively examined in the single-predictor and two-predictor settings. The question is whether the ability is enhanced or diminished after given multivariate information? The existing literature shows that linear combinations of individual forecasts can yield more precise predictions than the individual forecasts themselves. The underlying rationale for this is that such combinations have the capacity to diminish the volatility of the forecast while preserving the essential information that facilitates forecasting (Rapach et al., 2010).

Tracing back to the foundational work of Bates and Granger (1969), the forecast combination has been widely adopted in the literature (see Chen and Maung 2023, Rapach et al. 2010, and among others). Besides, the diffusion index, another multivariate approach, is also confirmed as a powerful tool for handling multivariate information (see, e.g., Wang et al. 2018). Motivated by prior literature, this section uses multivariate methods to explore the hypothesis that two-predictor predictive models incorporating an ESG market indicator can yield more precise return forecasts and superior economic value relative to single-predictor models that exclude the ESG market indicator.

The combined return forecast can be expressed as the weighted mean of individual forecasts. This encompasses forecasts derived from models that either include or exclude the ESG market indicator. Mathematically, this relationship can be described as:

$$\hat{R}^C_{t+1} = \sum_{i=1}^{N} \omega_{i,t}\hat{R}^i_{t+1} \tag{E.11}$$

where $\hat{R}^C_{t+1}$ is the combined return forecast at time t, $\omega_{i,t}$ is the weight assigned to the forecast from the ith individual predictive model, subject to the constraints that each weight is positive. Their sum equals one, $\sum_{i=1}^{N}\omega_{i,t} = 1$. The estimated individual forecast, $\hat{R}^i_{t+1}$, is obtained by running the single-predictor model $R_{t+1} = b_{i,0} + b_{i,1}C_{i,t} + u_{i,t+1}$, or the two-predictor model $R_{t+1} = c_{i,0} + c_{i,1}C_{i,t} + c_{i,2}ESG_{i,t} + uu_{i,t+1}$ for $i = 1, 2, \ldots N$. Here, $C_{i,t}$ corresponds to the variables delineated in Section 2.4 and Table A.2 in the appendix, and N specifies the total number of predictive models in consideration.

To determine the weights, we employ four prevalent methods: mean, DMSPE(L, 1), DMSPE(L, 0.9), and DMSPE(NonL). The first approach uses an equal-weighted mean of individual forecasts, wherein ω is set to $\frac{1}{N}$, in line with Claeskens et al. (2016) and Wang et al. (2019). The last three methods are rooted in the concept of discounted mean squared prediction errors (DMSFE), a development attributable to Stock and Watson (2004). The underlying principle is to assign higher weights to forecasts with lower MSFEs. Generally, the weight allocated to individual predictive model i is defined by

$$\omega_{i,t} = \frac{d_{i,t}^{-1}}{\sum_{i=1}^{N} d_{i,t}^{-1}} \tag{E.12}$$

where $d_{i,t}$ is the sum of the DMSFE values, expressed as:

$$d_{i,t} = \sum_{h=q+1}^{t} \sigma^{t-q}(R_h - \hat{R}_{i,h})^2 \tag{E.13}$$

Here, q is the length of the in-sample period, and σ is the discount factor. In accordance with Rapach et al. (2010), we set σ at values of 0.9 and 1. These choices lead us to name the methods as DMSPE(L, 1) and DMSPE(L, 0.9), respectively.

Additionally, the weight is also computed under a nonlinear weighting scheme:

$$\omega_{i,t} = \frac{exp(-\sum_{h=1}^{t-1}(R_h - \hat{R}_{i,h})^2)}{\sum_{i=1}^{N} exp(-\sum_{h=1}^{t-1}(R_h - \hat{R}_{i,h})^2)} \tag{E.14}$$

where the function exp refers to the exponential transformation. We name this method DMSPE(NonL).

The diffusion index approach employs principal components evaluated from a series of variables to estimate return forecasts. The diffusion index can be demonstrated as:

$$X_{i,t} = \xi_i' PC_t + z_{i,t} \tag{E.15}$$

Table E.6
Out-of-sample multivariate analysis.

	Single-predictor Regressions				Two-predictor Regressions			
	R^2_{OS}	*MSFE − adjusted*	CER Gain	Sharpe Ratio	R^2_{OS}	*MSFE − adjusted*	CER Gain	Sharpe Ratio
Mean	−1.438	−0.432	1.426	0.804	12.987***	2.625	18.069	1.503
DMSPE(L, 1)	−1.875	−0.431	2.214	0.837	12.594***	2.590	17.275	1.484
DMSPE(L, 0.9)	−1.412	−0.472	1.211	0.796	13.047***	2.627	17.886	1.485
DMSPE(NonL)	−1.444	−0.431	1.462	0.806	12.977***	2.624	18.035	1.503
DiffI	−1.623	−0.692	−1.894	0.691	13.219***	2.651	15.751	1.360

This table reports the out-of-sample performance using multivariate methods. *Mean*, *DMSPE*(*L*, 1), *DMSPE*(*L*, 0.9), and *dmspe*(*NonL*) are based on a combination forecast approach with different ways of weight allocation using Equation (E.12), (E.13), and (E.14), respectively. *DiffI* represents the diffusion index approach. Appendix E describes the multivariate analysis in detail. R^2_{OS}, introduced by Campbell and Thompson (2008), is the percentage reduction in mean squared forecast error of the predictive regression in comparison to the benchmark model, calculated according to Eq. (4). *MSFE − adjusted*, introduced by Clark and West (2007), is the statistics for testing hypothesis $H_0 : R^2_{OS} \leq 0$ against $H_A : R^2_{OS} > 0$. "CER Gain" is the difference of the certainty equivalent return (CER) between predictive models and the benchmark model. "Sharpe Ratio" is calculated as the annualized average excess return of a portfolio divided by its annualized standard deviation. See Section 4.2 for further details. The initial training period for the out-of-sample test is 120 observations. The data range from July 2007 to December 2021. ***, **, and * indicate significance at 1%, 5%, 10% levels, respectively.

where PC_t is a vector including the first q principal components. Here, we choose the first principal component extracted from our 20 variables listed in Table A.2 in the appendix, in line with Rapach and Zhou (2013). ξ_i' is the factor loading on variable i. The forecasts from predictive models that either include or exclude the ESG market indicator are given by:

$$\hat{R}^{DI}_{t+1} = \hat{a}^{DI}_1 + \hat{a}^{DI}_1 PC_t$$
$$\hat{R}^{DI}_{t+1} = \hat{b}^{DI}_1 + \hat{b}^{DI}_1 PC_t + \hat{b}^{DI}_2 ESG_t \quad \text{(E.16)}$$

Table E.6 reports the out-of-sample forecasting performance using multivariate techniques. The first four rows present results from forecast combinations with different weight distributions. The final method is rooted in the diffusion index approach. The most striking finding is the marked improvement in forecasting accuracy achieved through incorporating ESG data within combination strategies. All tests are statistically significant at the 1% level when the two-predictor regression includes the ESG market indicator. Moreover, the economic values shift substantially from a negative CER in the absence of ESG data to exceeding 17% across all methodologies when ESG factors are incorporated. The Sharpe ratios corroborate this observation. The Sharpe ratios further support this finding; when ESG factors are not considered, the Sharpe ratios range from 0.69 to 0.80. Conversely, these ratios climb to at least 1.36 when ESG variables are taken into account. Notably, the forecast based on an equal-weight combination yields the best results, thereby reinforcing the argument that simpler combination methods often outperform more sophisticated counterparts in out-of-sample prediction scenarios (Claeskens et al., 2016). In summary, the inclusion of the ESG indicator significantly enhances the predictive performance of combination approaches.

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